Dual-Band GNSS Antenna System with Integrated LNA for Autonomous Vehicle Applications

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Abstract

Accurate positioning in autonomous vehicles requires more than a basic GNSS antenna. Single-frequency designs are vulnerable to ionospheric delay and multipath interference, both of which are common in urban driving environments. This project addresses that gap by designing a compact dual-band GNSS front-end that operates across the L1 (1575.42 MHz) and L5 (1176.45 MHz) frequency bands.

The system uses two independently optimized microstrip patch antennas, a Wilkinson power combiner, a wideband BJT-based low-noise amplifier (LNA), and parallel third-order Chebyshev bandpass filters, all fabricated on FR-4 PCBs and designed in Keysight ADS. The LNA delivers greater than 18 dB gain across both bands. The Wilkinson combiner achieves a near-equal −3.4 dB power split with −22 dB port isolation. Each bandpass filter is centered on its respective GNSS band with greater than 30 dB out-of-band rejection. A bias tee circuit passes the RF signal with negligible attenuation while maintaining −140 dB RF-to-DC isolation.

Three PCBs were designed and ordered for physical fabrication. All subsystems met their target specifications in simulation, with VNA-based hardware validation planned as the final step.

Acknowledgments

The authors would like to thank Dr. Ali Attaran for proposing this project, providing access to simulation tools and lab resources, and offering technical guidance and feedback throughout the design process. His direction was instrumental in shaping both the system architecture and the direction of the design iterations.

The authors also thank Dr. Kenneth McIsaac for his input and support over the course of the project.

Nomenclature

Symbol	***Description***	***Units***
c	Speed of light in free space (3×10^8)	m/s
f_0	Center Frequency	Hz
f_1	L1 band frequency (1575 Mhz)	Hz
f_2	L5 band frequency (1176 Mhz)	Hz
W	Patch antenna width	mm
L	Patch antenna length	mm
W_g	Ground Plane width	mm
L_g	Ground Plane length	mm
H	Substrate thickness	mm
D	Inset feed depth	mm
S	Inset feed width	mm
ε_r	Relative permittivity of substrate	-
ε_{eff}	Effective dielectric constant	-
Z_{in}	Antenna input impedance	Ω
Z_o	System characteristic impedance	Ω
Z_b	Wilkinson branch impedance	Ω
λ_g	Guided wavelength	-
S11	Input reflection coefficient	dB
S21	Forward transmission coefficient	dB
S31	Reverse transmission coefficient	dB
S32	Port-to-port isolation coefficient	dB
S22	Output reflection coefficient	dB
X_C	Capacitive reactance	Ω
X_L	Inductive reactance	Ω
P_1	Pseudorange measurement at frequency f_1	m
P_2	Pseudorange measurement at frequency f_2	m
$P_{iono-free}$	Ionosphere-free pseudorange combination	m

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1 Introduction / Background

1.1 Problem Statement

Modern autonomous vehicles depend on precise, continuous positioning to make real-time navigation decisions. A positioning error of even a few dozens of centimeters can compromise lane-keeping, obstacle avoidance, and route planning. GNSS provides the primary source of absolute position, but the vehicle environment makes reliable reception genuinely difficult.

Typical off-the-shelf antennas are poorly suited to these conditions. Most are single-band, which leaves the system exposed to ionospheric delay errors that dual-frequency reception would otherwise cancel. They also lack integrated amplification, so cable and feed losses reduce system sensitivity directly. The outcome is a front-end that performs adequately in open-sky conditions but degrades in the environments where autonomous vehicles are most deployed: urban roads, parking structures, and suburban areas with overhead obstructions.

These constraints define the design challenge: a compact, vehicle-compatible GNSS RF front-end that operates across two frequency bands and includes sufficient low-noise amplification to maintain signal integrity under realistic deployment conditions.

1.2 Design Requirements

The design requirements for this project were initially defined during the proposal stage and later refined as the concept evaluation process moved the system toward a more practical dual-band GNSS front-end. At the highest level, the design was required to support operation at the GPS L1 frequency of 1575.42 MHz and the L5 frequency of 1176.45 MHz, since dual-band reception was central to improving positioning robustness over a single-frequency solution [1], [2], [8], [15]. The system also had to remain compatible with the available GNSS receiver, which required a single 50 Ω SMA-based RF output [3], [15], [18].

For the antenna subsystem, the original targets included a boresight gain of at least 3 dBic, an axial ratio below 3 dB, a VSWR no greater than 2:1 across each band, right-hand circular polarization, and a bandwidth of at least 20 MHz per band [4], [7], [8], [15]. However, the final concept evaluation showed that meeting all of these targets through a stacked circularly polarized structure would introduce excessive design complexity for the available timeframe and fabrication constraints [6], [10]. The final antenna requirement therefore shifted toward two

separately optimized rectangular microstrip patch antennas, one for L1 and one for L5, each designed for accurate resonance, 50 Ω feed compatibility, and acceptable impedance matching, while prioritizing manufacturability and ease of integration [4], [14], [15].

The RF front-end was required to provide sufficient gain and filtering to preserve extremely weak GNSS signals, which typically arrive below −120 dBm [15], [16], [19]. This led to the requirement for a wideband LNA covering both frequency bands, with a target gain of 20–30 dB, a low noise figure of 1.5 dB or better, and 50 Ω input and output impedance [16], [19]. As the design matured, the original 5 V supply assumption was replaced by a 3.3 V bias delivered through the coaxial output using a bias tee, in order to remain compatible with the selected receiver interface [16], [19].

Because the receiver accepted both GNSS bands through a single RF input, the front-end also required Wilkinson combiner and divider stages, separate L1 and L5 bandpass filtering, and a final recombination stage. The Wilkinson sections were designed in a 50 Ω environment with 70.7 Ω quarter-wave branches and a 100 Ω isolation resistor [16]. The filters were implemented as third-order Chebyshev bandpass networks with 0.5 dB passband ripple and 30 dB stopband attenuation. In the final design, the L1 filter used a 30 MHz passband with a 30 MHz transition band, while the L5 filter used a 24 MHz passband with a 25 MHz transition band [15], [16]. Since GHz-frequency lumped filters are strongly affected by parasitics, commercial SAW filters were also included as a practical backup option [15], [16].

From an implementation standpoint, the completed system was required to remain modular and manufacturable. This led to a three-board architecture consisting of separate L1 and L5 antenna PCBs and a dedicated RF signal-processing PCB interconnected through short 50 Ω SMA coaxial cables [3], [15], [18]. The RF board was implemented on a 2-layer FR-4 substrate using grounded coplanar waveguide routing, tapered transitions, and via stitching to maintain controlled impedance [11], [16]. Finally, each subsystem was required to be validated in simulation before integration, with the completed prototype prepared for subsequent VNA-based hardware testing [11], [15], [16], [19].

1.3 Detailed Literature Review

Global Navigation Satellite Systems (GNSS) support a wide range of applications, including automotive navigation, aviation, and high-accuracy positioning. In autonomous vehicle systems, the performance of the GNSS receiver depends strongly on both the antenna and the

receiver front-end, particularly in environments affected by weak signal strength, multipath interference, and electromagnetic noise [15]. Since autonomous vehicles require reliable and continuous positioning, the design of the antenna system must consider not only signal reception but also integration constraints, such as size, placement, and compatibility with the receiver front-end.

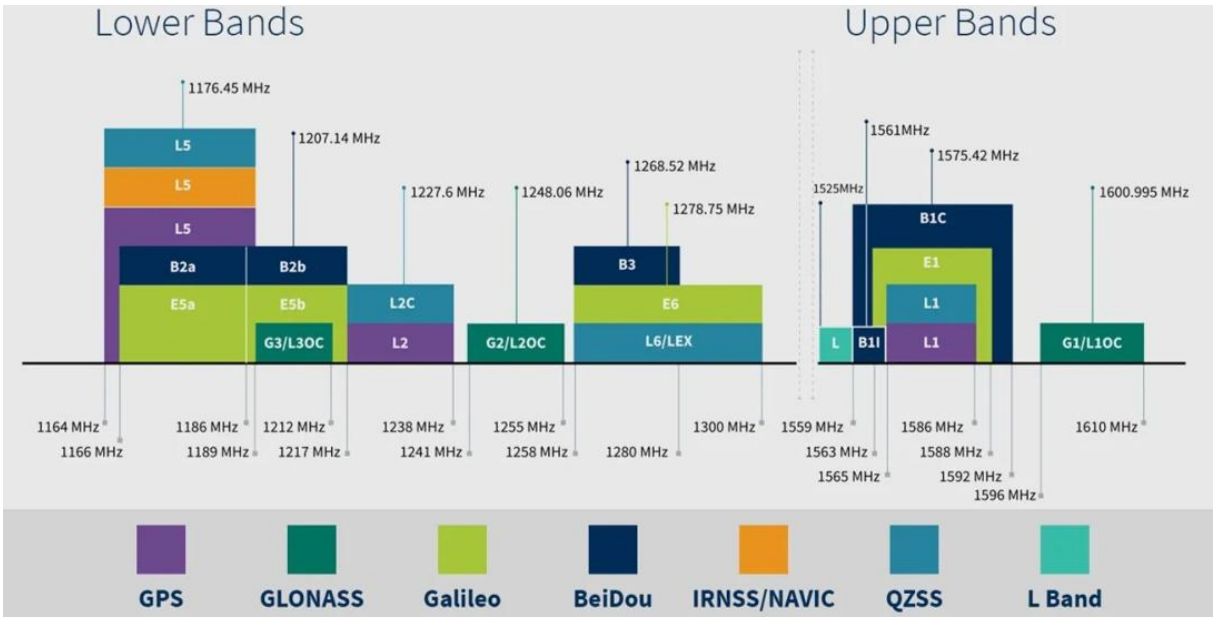


Figure 1.2.1 - GNSS Frequency Bands

While many earlier GNSS systems relied mainly on the L1 frequency band, more recent systems increasingly make use of multiple bands such as L1 and L5 to improve accuracy and robustness [1], [2]. Dual-band operation is especially useful because it allows the receiver to better compensate for ionospheric delay and improve positioning performance compared to single-frequency solutions [2], [8]. However, supporting both L1 and L5 also places stricter requirements on antenna bandwidth, impedance matching, and front-end signal conditioning.

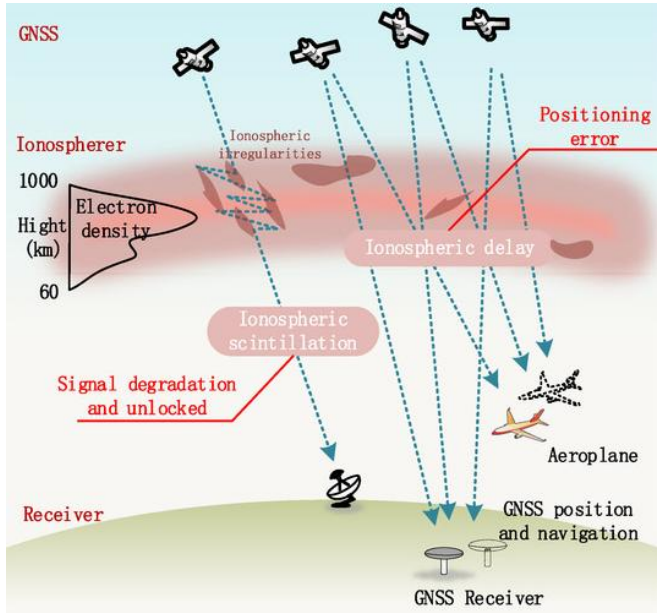


Figure 1.3.2 - Ionospheric Delay Illustration

Using dual-frequency signals (L1 and L5), the receiver can estimate and cancel out ionospheric delay because the error depends on frequency. Just by combining them with the ionosphere-free formula (1), the receiver reduces positioning error from ~3–5 m (L1 only) to ~1 m or better without external corrections.

$$P_{iono\text{-}free} = \frac{f_1^2 P_1 - f_2^2 P_2}{f_1^2 - f_2^2} \tag{1}$$

Several antenna types have been studied for GNSS applications, including helical antennas, dielectric resonator antennas, and microstrip patch antennas [4], [6]. Microstrip patch antennas are very popular in cars and other embedded systems because they are cheap, easy to integrate into PCBs, and have a low profile [4, 7]. Their main disadvantages are their narrow bandwidth and their sensitivity to ground plane size and nearby conductive structures [4], [7]. Even with these limitations, patch antennas remain a practical choice for compact vehicle-mounted systems, particularly when manufacturability and ease of integration are important design considerations.

An additional consideration in GNSS antenna design is polarization. Since GNSS satellite signals are right-hand circularly polarized, RHCP antennas are preferred because they reduce polarization mismatch loss [15]. More advanced dual-band designs, such as stacked patch

structures with hybrid couplers, can improve compactness and circular polarization performance [6], [10]. However, these approaches introduce greater design complexity, tighter fabrication tolerances, and more difficult tuning. For this reason, a simpler architecture using two separately optimized patch antennas can be more practical when balancing performance, manufacturability, and testing feasibility.

In addition to the antenna, the RF front-end has a major influence on overall system sensitivity. GNSS signals arrive at extremely low power levels, so a low-noise amplifier (LNA) is required to provide gain while adding as little noise as possible [16], [19]. Proper input and output matching, stable biasing, and compact RF layout are all important to maintaining good LNA performance at L-band frequencies [16], [19], making the LNA a critical part of the front-end design rather than simply an added gain stage.

Bandpass filters are also commonly used in GNSS front ends to reject out-of-band signals and reduce interference before the signal reaches the receiver [15], [16]. Lumped-element Chebyshev filters are attractive for their compactness and simplicity, but at GHz frequencies their response can be strongly affected by parasitics from real components and PCB layout. For this reason, practical GNSS designs may also rely on commercial SAW filters when more predictable high-frequency performance is needed.

Finally, RF interconnect design is an important part of the overall system. Wilkinson dividers and combiners are widely used because they provide equal power division, good impedance matching, and strong port isolation [16]. Grounded coplanar waveguide structures are commonly used on RF PCBs to maintain controlled impedance and reduce unwanted coupling [11].

Overall, the literature shows that reliable GNSS performance in autonomous vehicle applications requires coordinated design of the antenna, LNA, filtering, RF interconnects, and biasing network while also considering practical automotive integration constraints [3], [18].

1.4 Project Objectives

The objective of this project was to design a practical dual-band GNSS front-end for autonomous vehicle applications that could receive both the L1 and L5 frequency bands and improve positioning performance under weak-signal conditions. The system also needed to

remain suitable for PCB-based implementation and compatible with the GNSS receiver available for testing. As shown in Figure 2, the final architecture was refined into a signal chain consisting of two separately optimized microstrip patch antennas, a wideband low-noise amplifier, Wilkinson combiner and divider stages, bandpass filtering, and a receiver interface designed around a single coaxial output. (Figure 1.3.1).

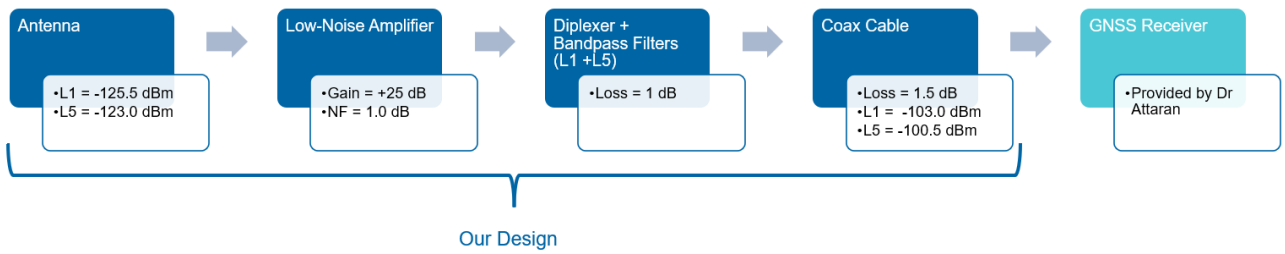


Figure 2 - Block diagram with numeric signal powers (dBm)

As the project developed, the design was refined into a system that includes Wilkinson combiner and divider stages, a grounded coplanar waveguide, and a bias tee to allow both RF and DC to share a single output connection.

The main goal was not only to design the individual subsystems, but also to ensure that they operated together as one complete RF front-end. This included tuning the antennas to the L1 and L5 bands, providing sufficient gain for weak GNSS signals, isolating the two bands through filtering, and recombining them into a single receiver-compatible output. As the design progressed, the architecture was further refined to include grounded coplanar waveguide routing and a bias tee so that RF and DC could share the same output path.

A further objective was to validate the design through simulation and prototype preparation. This involved using ADS to analyze each subsystem, finalizing PCB layouts for fabrication, and preparing the system for hardware testing. Overall, the project aimed to produce a technically feasible dual-band GNSS front-end that could support future physical validation and eventual vehicle integration.

2 Design Approach

2.1 Previous Iterations

The primary objective of this project involves designing a compact GNSS antenna for L1 and L5 frequency bands that can be integrated inside a vehicle while maximizing signal quality

and compactness. Multiple approaches were considered to accomplish this objective, and the first iteration of the system architecture is described as per figure 2.1.1.

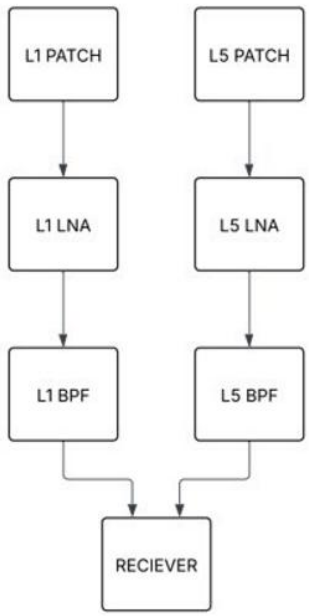


Figure 2.1.1 - Design Iteration One

The original concept used two separate microstrip patch antenna PCBs, one per band. Each PCB would use a 4-layer stack-up with a dedicated low-noise amplifier and bandpass filter on the bottom layer to maximize compactness. This approach was favored early in the design process because it allowed each antenna to be independently optimized for bandwidth, impedance matching, and overall performance.

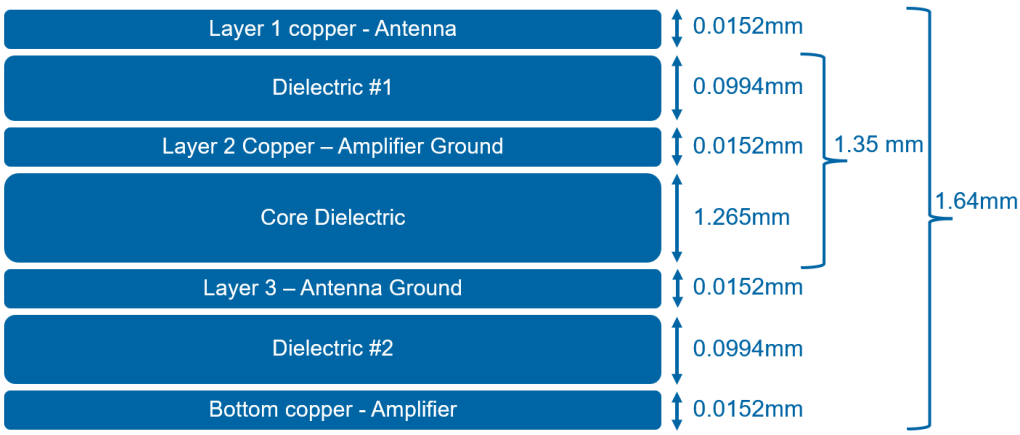


Figure 2.1.2 – 4-Layer PCB Stack-up Design for Iteration One

The stack-up shown in Figure 2.1.2 uses the thickest core dielectric available for 1.6 mm 4-layer boards at the intended manufacturer, JLCPCB. A thicker board was not pursued due to the exponential increase in manufacturing costs beyond 1.6 mm, which rises from approximately $7 to over $100 for a 2 mm board. The position of the amplifier's and antenna's grounds were inversed so that the substrate would be maximized for the antenna, since substrate thickness is proportional to the bandwidth of patch antennas and delivers better performance the greater it is.

3) JLC04161H-3313 Stackup

Layer	Material Type	Thickness	
Top Layer	Copper	0.035mm	
Prepreg	3313*1	0.09940mm	
Inner Layer L2	Copper	0.0152mm	1.3mm H/HOZ with copper
Core>	Core	1.265mm	
Inner Layer L3	Copper	0.0152mm	
Prepreg	3313*1	0.09940mm	
Bottom Layer	Copper	0.035mm	

Figure 2.1.3- JLPCB 4-Layer Stack-up Option with the Thickest Core Dielectric

This approach was recommended by project advisor Dr. Attaran, as it closely mirrors commercial GNSS patch antenna designs, where signal processing elements are integrated directly on the same board as the antenna. This level of integration has become standard practice given the extremely low received power levels of GNSS signals, typically around −120 dBm.

However, several challenges led the team to reconsider this approach. The design would have required blind vias, a tightly constrained layout accommodating the LNA, bandpass filter, and bias tee, and a combined output within a limited board area, all of which would have significantly complicated both fabrication and simulation. A further limitation was that the feed method produced linear polarization, introducing a polarization mismatch with GNSS satellite signals, which require RHCP and could lose up to 3dB of system gain. Given these constraints, the team decided to explore alternative design approaches.

To resolve the polarization issue of the design, the 4-layer stack up was modified to house both antenna patches on the same PCB, to use a coaxial feed instead of a microstrip edge feed and to have the signal processing components (LNA, BPFs, etc..) on a separate board.

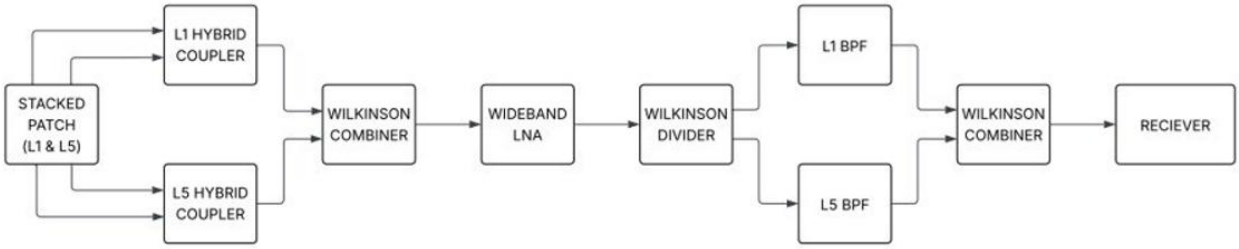


Figure 2.1.4- Design Iteration Two

This would involve two orthogonal probes for each patch as per Figure 2.1.5 to maintain circular polarization through a circuit called a hybrid coupler as per Figure 2.1.7. The benefit of this approach is that it maximizes right hand circular polarization through a coupling circuit, as well as substrate thickness for the L5 patch (Figure 2.1.7). The problem is that it introduces significantly more complexity to the project at the front end and requires a far more precise finish. After reviewing multiple approaches from published papers and attempting to implement, the team felt stuck as we did not know which approach was the best and would be feasible to implement due to its complexity.

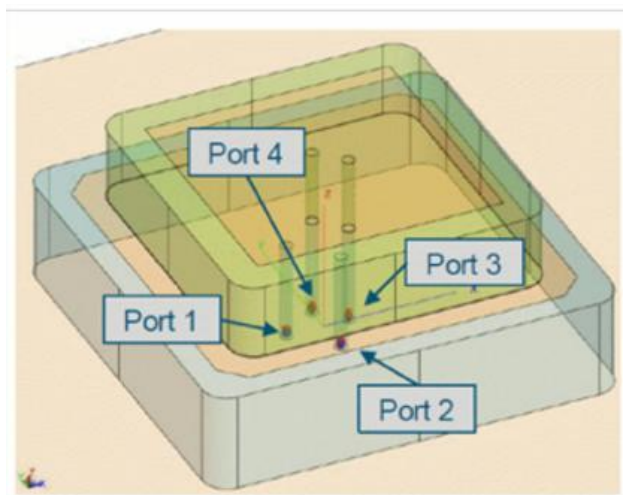


Figure 2.1.5- Stacked Patch Antenna Model [16]

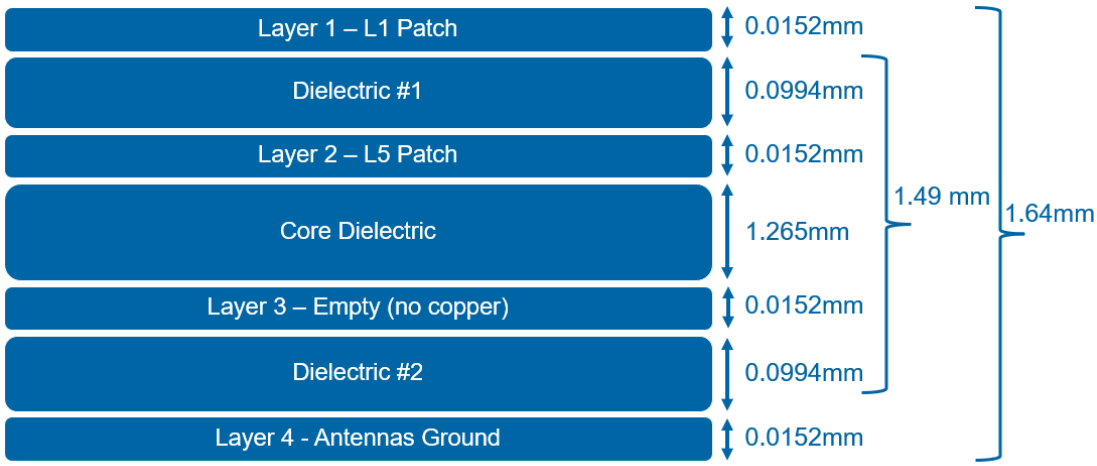


Figure 2.1.6- 4-Layer PCB Stack-up Design for Iteration Two

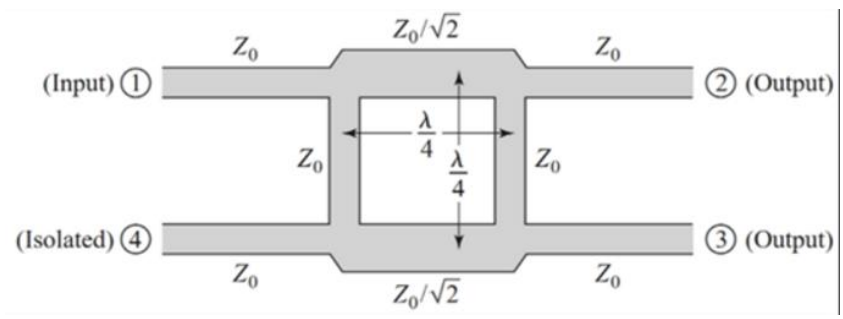


Figure 2.1.7 - Hybrid Coupler Circuit [10]

2.2 Concept Evaluation and Final Concept Selection

The final design iteration was developed after having a meeting with our advisor and discussing all our design iterations and how we should move forward. It involved taking concepts from each main design to find a balance between how compact the prototype would be, its complexity, cost, and testability in Keysight ADS EM simulations.

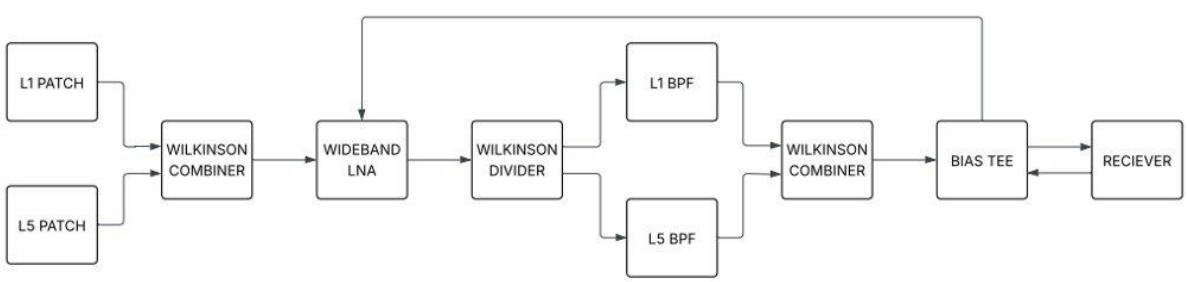


Figure 2.2.1 - Final System Architecture

The final system architecture for the dual-band GNSS consists of 3 separate PCBs that connect to each other using SMA cables to allow for isolated performance validation and modularity. Two separate antenna PCBs connect to the third PCB using a Wilkinson combiner and feed into a wideband LNA. The use of a single wideband LNA was preferred to reduce component count, board space, and complexity, at the cost of independent narrowband amplification.

The output of the LNA is fed into a Wilkinson power divider to split the signal so that L1 and L5 can each be refined by their own bandpass filter. This stage ensures strong isolation of the desired frequencies and helps eliminate out-of-band interference. The filtered signals are then recombined using a third Wilkinson power combiner and passed into a bias tee, supplying 3.3 V bias to the LNA while RF is injected from the system into the receiver.

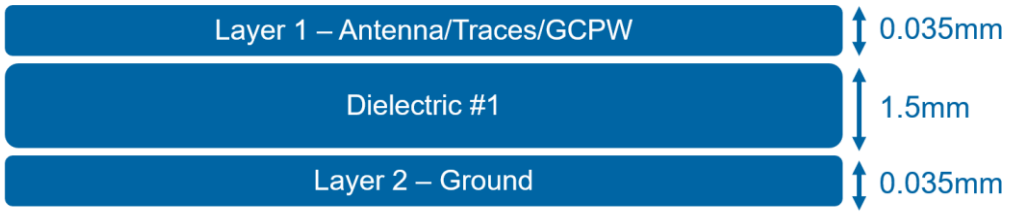


Figure 2.2.2 - Final Iteration Stack-up

Another benefit of this architecture is the simplicity of the stack-up for all the boards, which is now only 2-layers as per figure 2.2.2.

This method is realized in two variants: one using commercial SAW bandpass filters and one using equivalent lumped components. Since lumped components are not feasible at the L1 and L5 frequencies, purchased components are used to validate the design. The third implementation places all stages on a single PCB, combining the patch antennas and RF signal processing into the most compact form, though at the greatest design complexity and risk. To further support validation, a modular RF signal processing PCB was also designed with SMA pads between each stage, allowing individual sections to be physically separated and tested in isolation before being evaluated as a complete cascaded system.

3 Design Analysis

3.1 Engineering Techniques/Software tools

The main software tool used for this project was Keysight Advanced Design System (ADS). ADS was chosen because it allowed both circuit-level RF design and layout-based EM simulation to be completed within the same environment. This was important for this project since the final system included antennas, transmission lines, filters, an LNA, and passive RF routing stages that all had to work together as one front-end. ADS was used to design and simulate the wideband LNA, the L1 and L5 bandpass filters, the bias tee, and the overall cascaded RF signal chain.

For the antenna subsystem, standard microstrip patch antenna equations were first used to calculate the initial dimensions of the L1 and L5 patches. These values provided a starting point based on substrate properties and operating frequency. From there, ADS was used to iteratively tune the patch length and inset-feed dimensions until the desired resonant frequencies and return-loss levels were achieved. This combination of first-pass calculations and simulation was necessary because the final antenna response was sensitive to feed geometry, substrate thickness, and fabrication tolerances.

For the RF signal-processing board, ADS LineCalc was used to determine the required dimensions for the grounded coplanar waveguide traces and Wilkinson quarter-wave sections. Since these values are only first-order estimates, Momentum EM simulation was then used to evaluate the practical PCB layouts, especially for the Wilkinson combiner/divider and other distributed RF structures. This was important because PCB-level effects such as discontinuities, grounding, and trace geometry can significantly affect performance at GNSS frequencies.

Other engineering techniques used throughout the design included S-parameter analysis, Smith chart interpretation, and PCB layout verification. These were used to evaluate impedance matching, gain, insertion loss, port isolation, and manufacturability across the different stages of the design. Overall, the selected tools provided a consistent workflow from initial calculations to final simulated PCB implementation.

3.2 Final Analysis/Calculations

Final analysis of the proposed GNSS system above was conducted using a combination of EM simulations, RF system analysis, and theoretical design calculations. The main tool used to verify the feasibility of our design is Keysight's Advanced Design System (ADS) software. Each component of the system was tested individually to ensure it functioned properly and met its design specifications. Once completed, each stage was cascaded together to verify the entire system was functioning as intended.

3.2.1 L1 and L5 Patch Antenna

The initial dimensions of the rectangular microstrip patch antennas were calculated using standard design equations [12]. The patch width W was obtained from

$$\mathrm{W} = \frac{c}{2f_0\sqrt{\frac{\varepsilon_r+1}{2}}} \tag{2}$$

where f_0 is the band frequency and ε_r is the relative permittivity of the substrate.

The Effective Dielectric Constant accounts for fringing effects and is given by:

$$\varepsilon_{eff} = \frac{\varepsilon_R+1}{2} + \frac{\varepsilon_R-1}{2} + \left[\frac{1}{\sqrt{1+12\left(\frac{h}{W}\right)}}\right] \tag{3}$$

where h is the substrate thickness

Length:

$$L = \frac{c}{2f_0\sqrt{\frac{\varepsilon_R+1}{2}}} - 0.824h\left(\frac{(\varepsilon_{eff}+3)\left(\frac{W}{h}+0.264\right)}{(\varepsilon_{eff}-2.58)\left(\frac{W}{h}+0.8\right)}\right) \tag{4}$$

Input Impedance:

$$Z_{in} = \frac{90}{\varepsilon_r}\left(\frac{\varepsilon_r+1}{2}\right)^2\frac{L}{W} \tag{5}$$

Ground Plane Width:

$$W_g = W + 6h \tag{6}$$

Ground Plane Length:

$$L_g = L + 6h \tag{7}$$

Figure 1 below illustrates the edge-fed rectangular patch antenna configuration used in this work.

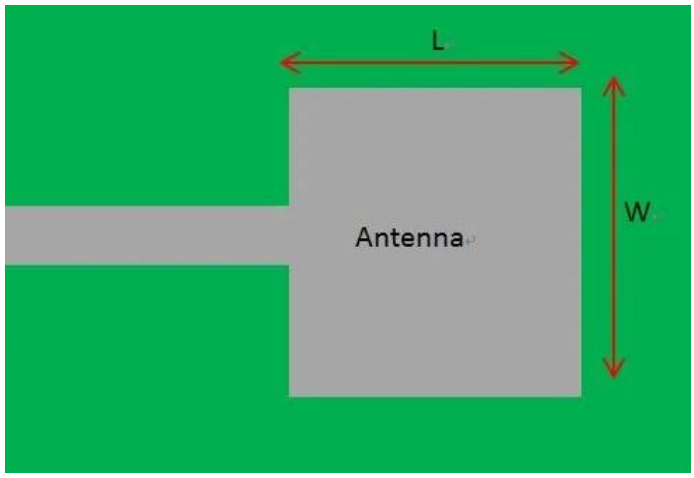


Figure 3.2.1 – Edge-Fed Patch Antenna

The substrate dielectric constant and thickness were selected based on the JLCPCB stack-up option with the thickest core dielectric. Initially, our concept was to use a 4-layer stack-up but since switching to a 2-layer one, the substrate thickness naturally increased. The calculation results in table 1 and 2 were updated since the midterm progress report to match the final concept chosen.

	Know Values				
	Substrate ε_r	Substrate Thickness (in mm)	Frequency (in MHz)	Feedline Impedance Z_0 (in Ohms)	
	4.4	1.5	1575.42	50	
Calculations					
Input Impedance (Ohms)	Bandwidth (MHz)	Width (mm)	Length (mm)	Ground Width (mm)	Ground Length (mm)
295	10.8	57.912	43.942	66.912	52.942

Table 3.2.1 – L1 Patch Antenna Preliminary Calculations

The L5 band requires a wider bandwidth than L1, which is especially challenging because of its lower center frequency. Since bandwidth in patch antennas is proportional to substrate thickness, a standard 1.6 mm PCB was found to be insufficient to cover the required 20 MHz bandwidth. Thus, two single-layer PCBs were mechanically stacked to achieve an effective substrate thickness of 2.9 mm, enabling the antenna to meet the L5 bandwidth requirements.

To achieve this, we found an edge-mounted SMA connector capable of holding two boards together (Digi-Key part number: 1868-1198-ND) as per figure 3.2.2.

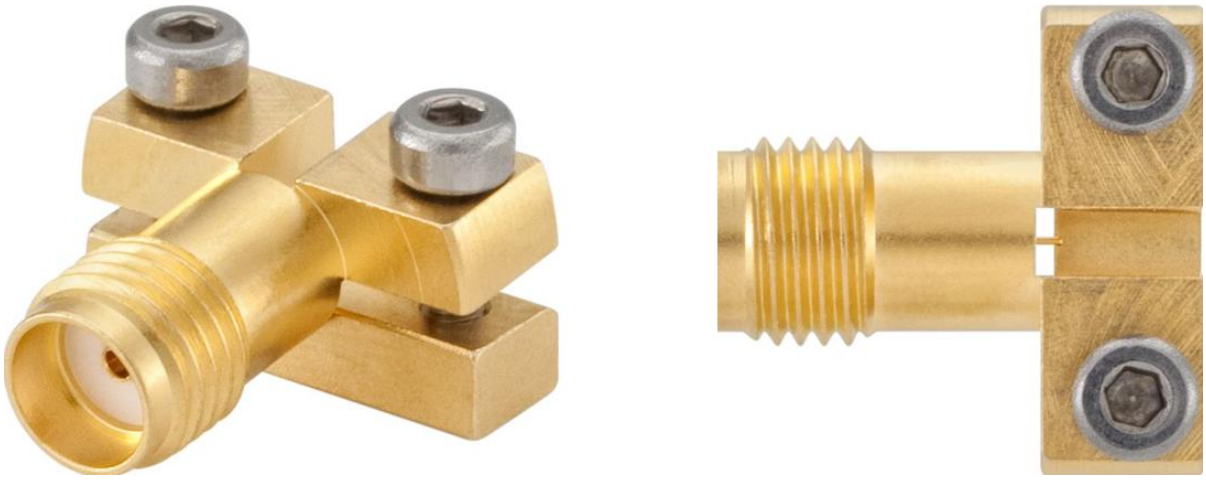


Figure 3.2.2 - SMA Connector for L5

	Know Values				
	Substrate ε_r	Substrate Thickness (in mm)	Frequency (in MHz)	Feedline Impedance Z_0 (in Ohms)	
	4.4	2.9	1176.45	50	
Calculations					
Input Impedance (Ohms)	Bandwidth (MHz)	Width (mm)	Length (mm)	Ground Width (mm)	Ground Length (mm)
300	5.99	77.47	59.436	94.87	76.836

Table 3.2.2 – L5 Patch Antenna Preliminary Calculations

A major challenge in microstrip patch antenna design is matching the 50 Ω feedline impedance. Several matching techniques are available, but the **inset-fed method** was selected due to its simplicity and ease of fabrication.

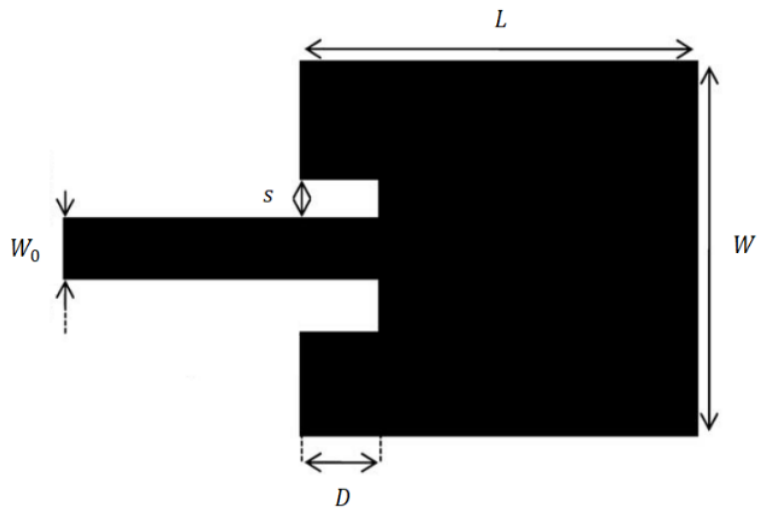


Figure 3.2.3 Patch Antenna with Inset Feed

Initial simulations performed in Keysight ADS using the analytically derived dimensions did not meet the required performance targets. As a result, an iterative optimization process was carried out by adjusting the patch length L, inset depth D, and inset width S.

For the first prototype iteration, 10 steps of iterative optimization were required to achieve a return loss better than −25 dB at the target frequency. The corresponding optimization steps and results are summarized in Table 3.

Iteration #	**L (mm)**	**D (mm)**	**S (mm)**	**Center f (GHz)**	**Peak Return Loss (dB)**	**Impedance (50 Ohms)**
1	43.18	11.8	3.76	1.6	-3.76	Not Measured
2	43.7	11.8	3.76	1.6	-12.78	Not Measured
3	44.2	11.8	3.76	1.5875	-18.58	Not Measured
4	44.4	11.8	3.76	1.825	-18.6	Not Measured
5	44.6	11.8	3.76	1.575	-20.18	1.103 + j0.179
6	44.6	11.6	3.76	1.575	-21.86	1.085 + j0.140
7	44.6	11.6	3.56	1.574	-23.77	1.076 + j0.111
8	44.5	11.4	3.56	1.577	-25.03	1.095 + 0.070
9	44.6	11.4	3.56	1.574	-26.14	Not Measured
10	44.57	11.3	3.56	1.575	-27.25	1.066 + j 0.061

Table 3.2.3 – Iterative Optimization of L1 Patch for Concept Iteration 1

Since then, the design changed from a 4-layer to a 2-layer stack-up with a thicker substrate, so the antenna dimensions needed to be slightly adjusted and were further optimized as per Table 4.

Iteration #	L (mm)	D (mm)	S (mm)	Center f (GHz)	Peak Return Loss (dB)	Impedance (50 Ohms)
1	43.942	11.8	3.62	1.5862	-19.7	1.136 + j0.175
2	44.2	11.8	3.62	1.5775	-20.7	1.051 + j0.207
3	44.25	12.5	3.62	1.575	-16.42	0.999 + j0.435
4	44.25	11.1	3.62	1.5762	-29.13	1.060 + j0.040
5	44.3	11.1	3.62	1.575	-30.03	1.035 + j 0.054

Table 3.2.4 – Iterative Optimization of L1 Patch for Final Concept

Similarly, the L5 antenna required 12 optimization iterations to meet the same return loss criterion for the old 4-layer concept.

Iteration #	L (mm)	D (mm)	S (mm)	Center f (GHz)	Peak Return Loss (dB)	Impedance (50 Ohms)
1	59.44	21.7	3.76	1.15	-5.28	Not Measured
2	59	15	4.76	1.165	-17.2	Not Measured
3	58.6	15	5.26	1.175	-17.3	1.135 + j0.261
4	58.6	14.5	5.26	1.175	-19	1.188+j0.159
5	58.6	14	5.76	1.175	-15.95	Not Measured
6	58.6	14	5.26	1.175	-21.4	Not Measured
7	58.6	13.8	5.26	1.1765	-24.95	1.024 + j0.112
8	58.6	13.8	5.36	1.1765	-24.68	1.031 + j0.1145
9	58.6	13.8	5.16	1.176	-26.5	1.022 + j0.0933
10	58.5	13.8	5.06	1.1775	-27.02	1.026 + j0.0865
11	58.575	13.8	5.06	1.176	-26.64	1.056 + j0.774
12	58.57	13.8	5.06	1.1765	-26.44	0.994 + j0.095

Table 3.2.5 – Iterative Optimization of L5 Patch for Concept Iteration 1

It then required 5 more iterations for the final concept with the 2 mechanically stacked boards as per Table 6.

Iteration #	L (mm)	D (mm)	S (mm)	Center f (GHz)	Peak Return Loss (dB)	Impedance (50 Ohms)	Bandwidth (MHz)
1	64.3	14.8	4.32	1.1625	-23.68	0.906 - j0.082	23.75
2	63.3	14.8	4.42	1.177	-26.2	0.928 - j0.062	23.75
3	63.4	14.8	4.52	1.17625	-26.43	0.935 - j0.066	23.75
4	63.5	14.9	4.52	1.175	-26.97	0.929 - j0.050	25
5	63.5	15	4.52	1.175	-27.54	0.931 - j0.042	25

Table 3.2.6 – Iterative Optimization of L5 Patch for Final Concept

The simulations were made by combining a wide sweep, a mid sweep and a narrow sweep as per figure 3.2.4.

	Type	Fstart	Fstop	Npts	Step	Enabled
1	Linear	1.125 GHz	1.225 GHz	21	0.005 GHz	☑
2	Linear	0.5 GHz	2 GHz	16	0.1 GHz	☑
3	Linear	1.15 GHz	1.2 GHz	41	0.00125 GHz	☑

Figure 3.2.4 - Frequency Simulation Plans for L5

Figure 4.1.4– JLPCB Gerber View of L1 (left) and L5 (right) antenna PCBs (Not to scale)

3.2.2 PCB and Transmission Lines

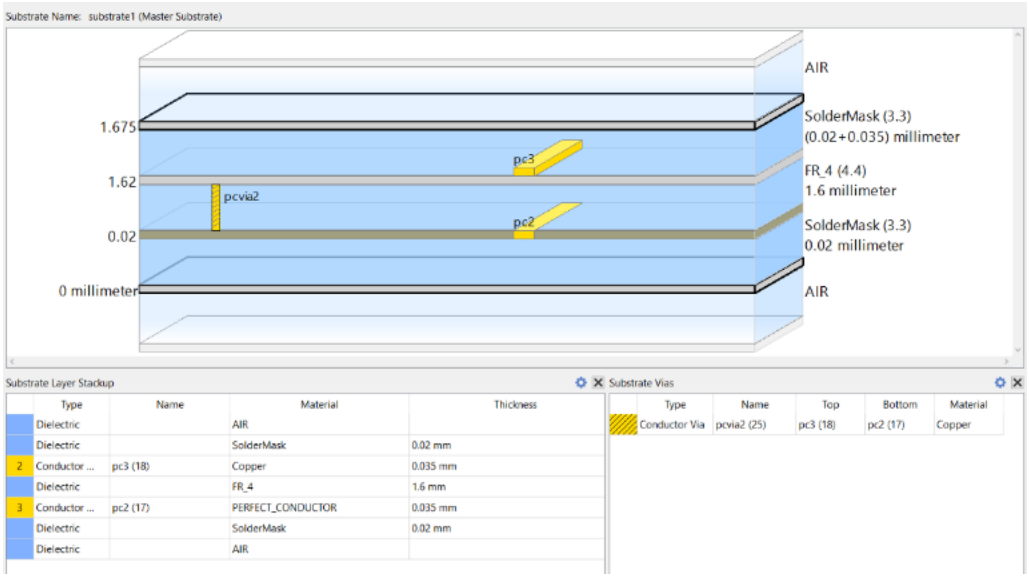


Figure 3.2.2.1 – RF Signal Processing Board PCB Substrate

The PCB substrate was defined using a standard 2-layer stack up based on FR-4 with a thickness of 1.6 mm and relative permittivity of 4.4. Copper layers of 35 µm thickness were used for both the top signal layer and bottom ground plane. Thin solder mask layers (0.02 mm) were included above and below the substrate to more accurately model practical fabrication conditions. The bottom layer serves as a continuous ground plane, while vias connect the top ground regions to this plane, forming a grounded coplanar waveguide (GCPW) structure. This stack up provides a good balance between cost, manufacturability, and acceptable RF performance at GNSS frequencies.

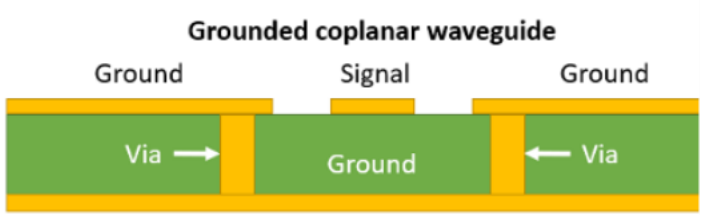


Figure 3.2.2.2– RF Signal Processing Board Transmission Line Type – GCPW [11]

A GCPW structure was selected for all RF traces in the signal processing PCB to provide controlled impedance, strong field confinement, and improved shielding from external

interference. As shown in Figure 3.2.2.2, the signal trace is surrounded by ground on the same layer, and via stitching connects it to the bottom ground plane along the signal line. A small gap clearance of 0.3 mm was selected for the above in accordance with JLPCB manufacturing tolerances to minimize trace width since the parameters are proportional.

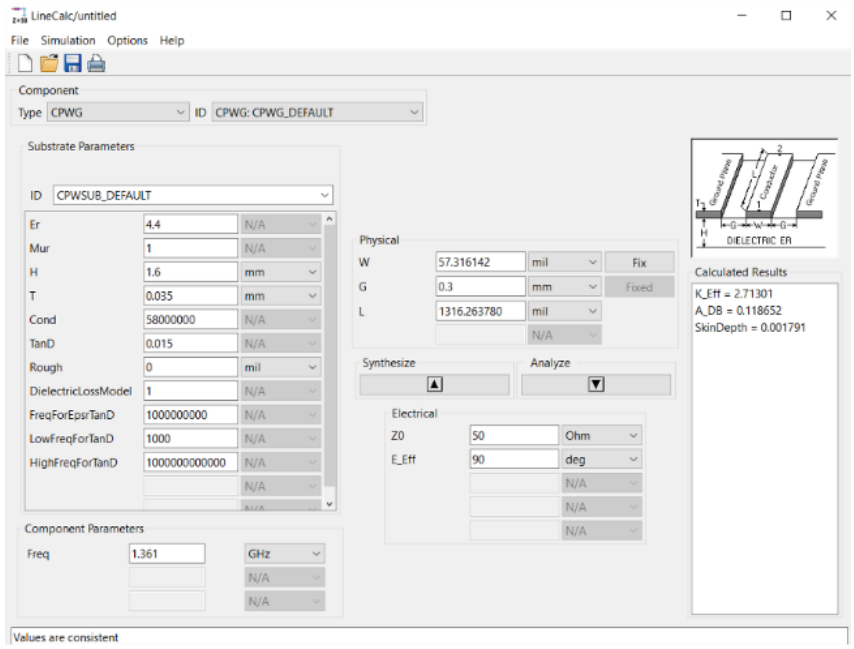


Figure 3.2.2.3– GCPW Line Calc for 50ohm Lines

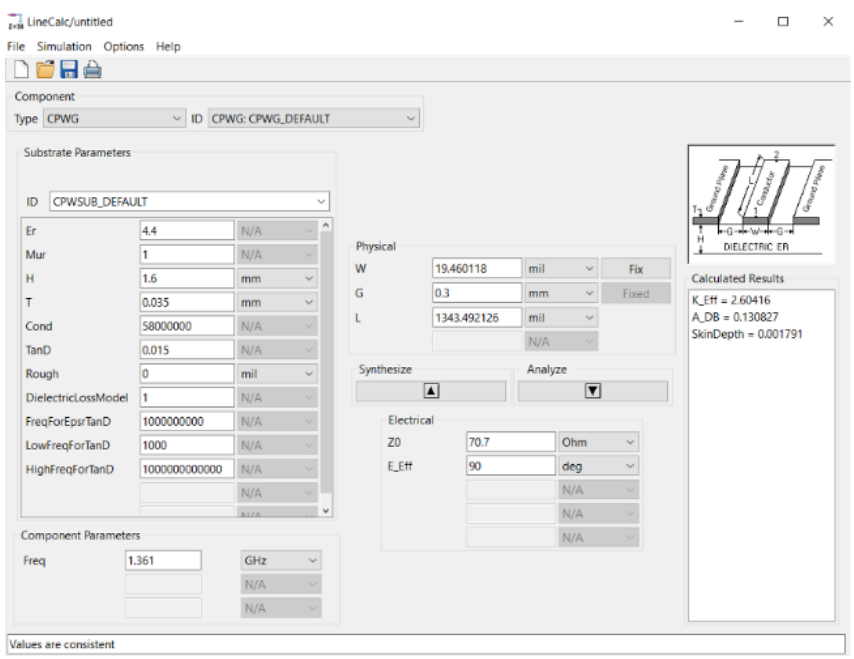


Figure 3.2.2.4– GCPW Line Calc for 70.7ohm Lines (Wilkinson)

Using ADS LineCalc, the CPWG dimensions were synthesized for a 50 Ω characteristic impedance at the geometric center frequency of L1 and L5 (1.36 GHz). The resulting dimensions were:

- Trace width: W = 1.46mm @ 50Ω (Standard System Transmission Line)
- Trace width: W = 0.49mm @ 70.7Ω (Wilkinson Quater Wave Section)
- Clearance: G = 0.3 mm

All RF sections were connected using short transmission line segments to minimize additional loss and unwanted phase shifts. Where transitions occurred between different trace widths (SMD component pads), tapered traces were implemented instead of having sharp connections. These tapers provide a gradual impedance transition, reducing reflections and improving overall matching across the system. This approach is especially important at GNSS frequencies, where even small discontinuities can degrade performance.

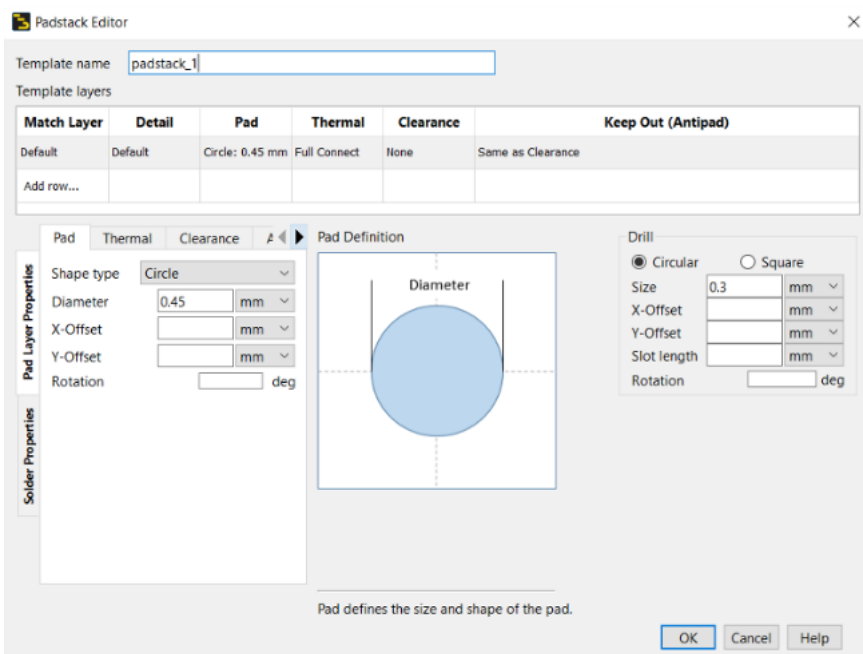


Figure 3.2.2.5 – Via Definition

Via stitching was implemented along all RF traces to suppress parasitic effects and improve ground continuity across lines. Based on guided wavelength calculations, via spacing was selected within the recommended range of λg/20 to λg/10 where $\lambda g = \frac{c}{f\sqrt{Eeff}}$. A more compact stitching distance of 2-4mm center to center was used, which is smaller than calculated

for better results. Heavier via placement was required to improve grounding at key junctions like the Wilkinson power splitter and SMA pads where jacks are mounted. Vias were implemented with a diameter of approximately 0.3 mm in accordance with JLPCB specifications, balancing a low inductance return path with manufacturability.

3.2.3 Wilkinson Power Divider and Combiner

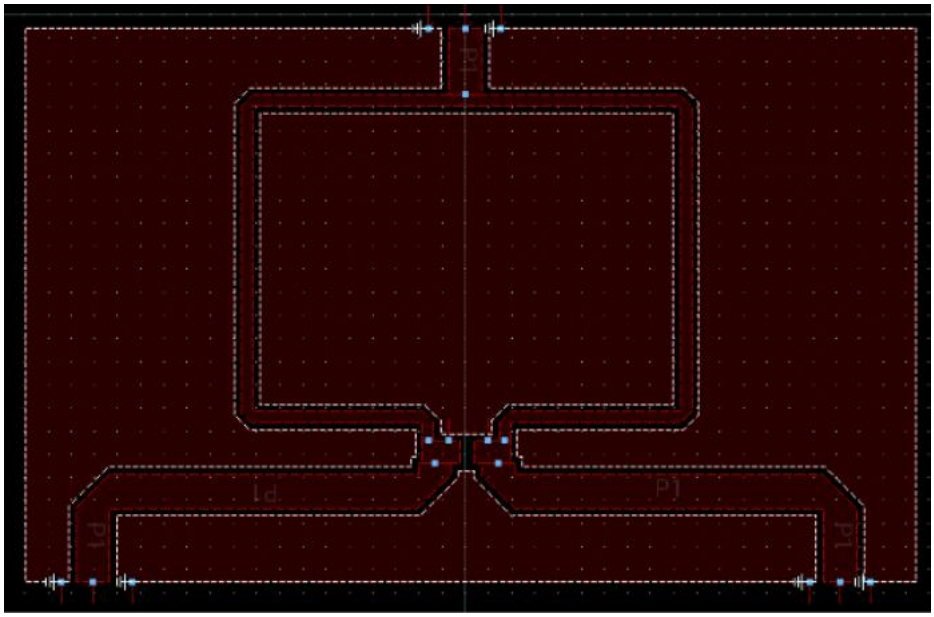


Figure 3.2.3.1 – Wilkinson Layout Design

S-parameter Ports

Number	Gnd Layer	Name	Feed Type	Ref Impedance [Ohm]
1	N/A	P1	Auto	50 + 0i
P1				
P6				
P7				
2	N/A	P2	Auto	50 + 0i
P2				
P8				
P9				
3	N/A	P3	Auto	50 + 0i
P3				
P10				
P11				
4	N/A	P4	Auto	100 + 0i
P4				
P5				

Figure 3.2.3.2 – Port Setup for EM Simulations

A Wilkinson power divider/combiner was designed to equally split L1/L5 signals while maintaining strong impedance matching. An isolation resistor was added to deal with any mismatched power between ports.

- System impedance Zo = 50Ω
    - Line Calc Trace width = 1.46mm
- Branch impedance Zb = √2Zo= 70.7Ω
    - Line Calc Trace width = 0.49mm
- Isolation Resistor R = 2Zo = 100Ω

The design was centered at geometric centre frequency 1.361 GHz, selected to provide operation across both GNSS bands. The electrical length of each branch is approximately a quarter wavelength section based on the frequency where $\lambda g = \frac{c}{4f\sqrt{Eeff}}$.. Using this equation and line calc tool, the quarter wavelength of each branch was calculated to be approximately 34.1mm. Final tuning of the branch sections was required in ADS momentum simulation to account for layout effects, effective permittivity, and substrate.

3.2.4 Wideband LNA

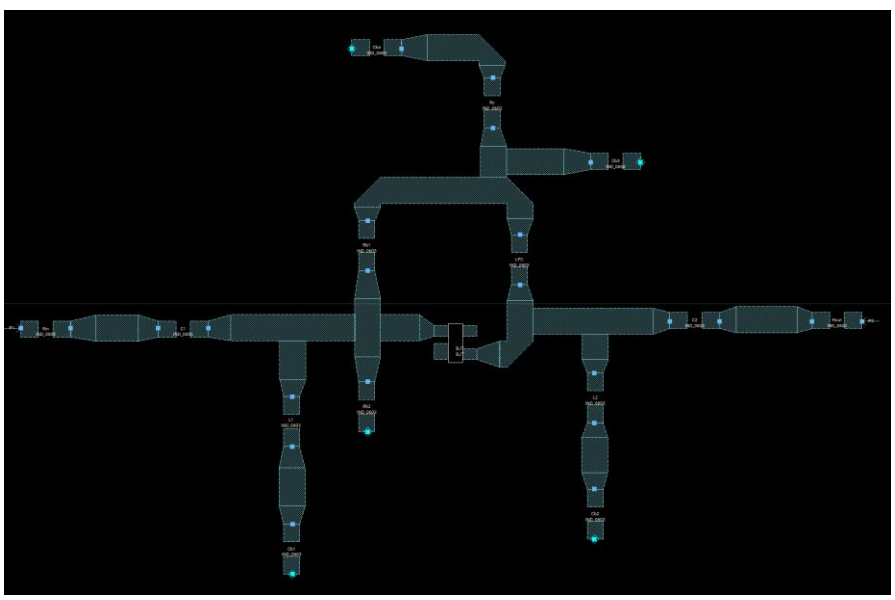


Figure 3.2.4.1 – Wideband LNA ADS Layout

The layout of the wideband low-noise amplifier (LNA) was implemented in Keysight ADS following challenges encountered when attempting to directly translate the schematic into a manufacturable PCB design. In particular, discrepancies between ideal schematic components and the physical dimensions of commercially available surface-mount devices required the layout to be adjusted to ensure compatibility with real components.

The LNA plays a critical role in the GNSS receiver front-end, as it is responsible for amplifying extremely weak incoming signals from satellites while introducing minimal additional noise. Since GNSS signals at the L1 (1.575 GHz) and L5 (1.176 GHz) bands are typically below −120 dBm upon reception, the performance of the LNA directly determines the overall sensitivity and signal-to-noise ratio of the system.

The layout is designed to support wideband operation across both frequency bands, which requires careful impedance matching at the input and output, as well as stability across the operating range. Microstrip transmission lines and matching networks are used to achieve a 50 Ω impedance environment, minimizing reflections and ensuring maximum power transfer. Additionally, the layout geometry has been kept as compact and symmetric as possible to reduce parasitic inductance and capacitance, which can significantly impact RF performance at these frequencies.

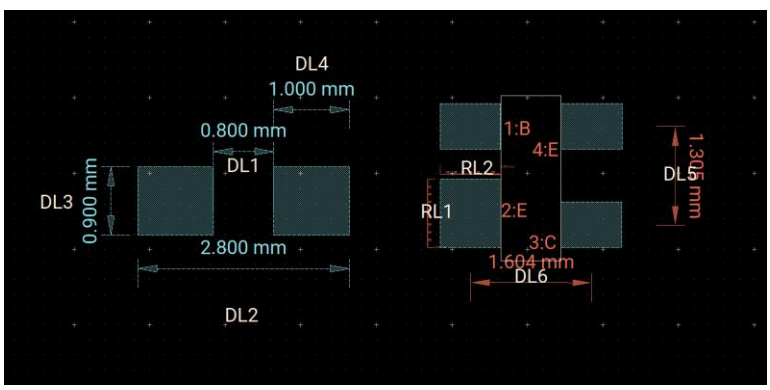


Figure 3.2.4.2 – Component dimensions

Figure 3.2.4.2 shows the physical dimensions used for the surface-mount components in the LNA design. All passive components were selected in the 0603 (1608 metric) package to maintain consistency and simplify PCB layout and assembly.

3.2.5 L1 and L5 Bandpass Filters

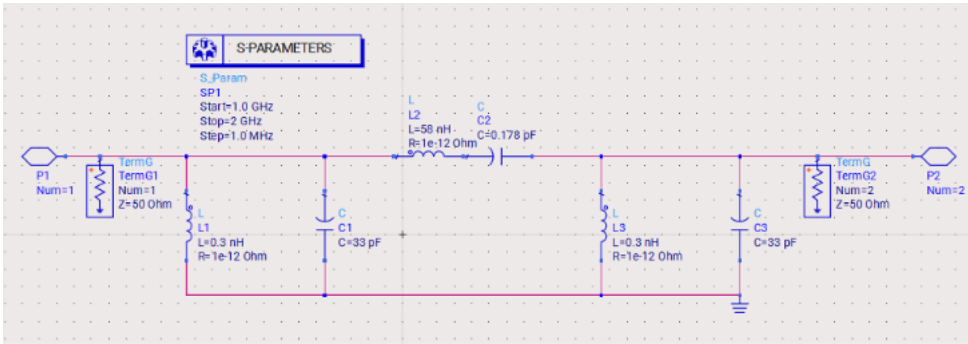


Figure 3.2.5.1 – L1 Bandpass Filter Circuit Schematic

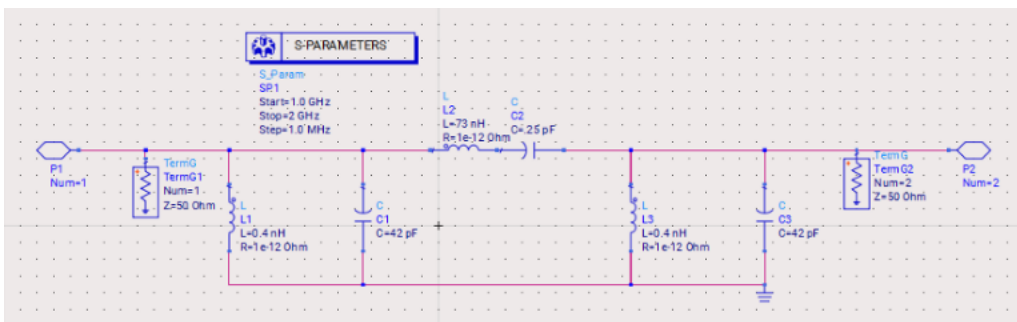


Figure 3.2.5.2– L5 Bandpass Filter Circuit Schematic

Two third order bandpass filters were designed for L1 and L5 using ADS filter designer. A Chebyshev filter topology was selected to achieve a sharper transition band for out of band rejection and to minimize filter order at the cost of a small passband ripple. The filters were designed with the following parameters:

- Order: 3
- Passband Ripple: 0.5dB
- Stopband Attenuation: 30dB

These specifications provided the following passband results:

- Centered passband at1.575GHz and 1.176GHz respectively
- 30MHz passband with sharp 30MHz transition band
- Negligible 0-1dB insertion loss
- 50-70dB attenuation outside the passband

While the lumped element Chebyshev filters provide good theoretical performance, practical limitations exist at GHz frequencies due to parasitic effects in small inductors and capacitors. These parasitic can degrade filter response and shift the center frequency. To mitigate this risk, commercial SAW filters were selected as a backup, providing a more stable and reliable filtering solution if measured performance of the lumped filters is insufficient. The SAW filter selected provides similar results to the LC model simulated validating overall system results and pads for the PCB were drawn accordingly to product specification sheet as per Appendix 9.1.

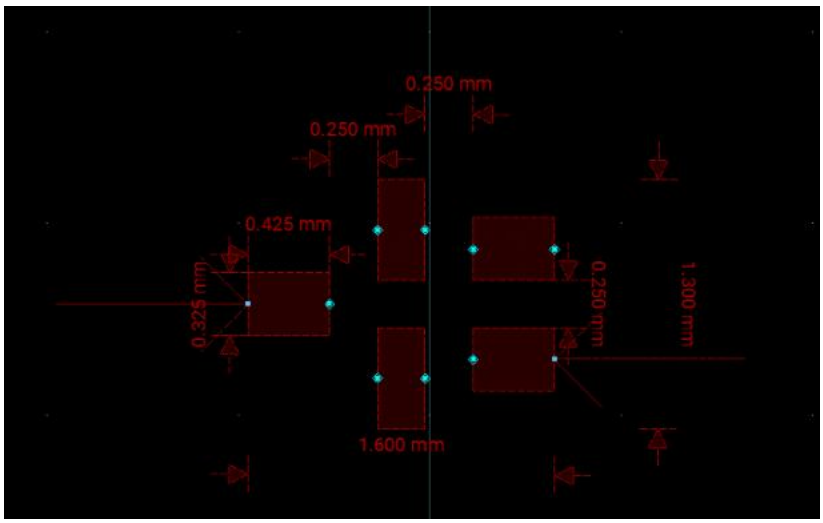


Figure 3.2.5.3 – L1 and L5 Pad Layout for SAW Filter

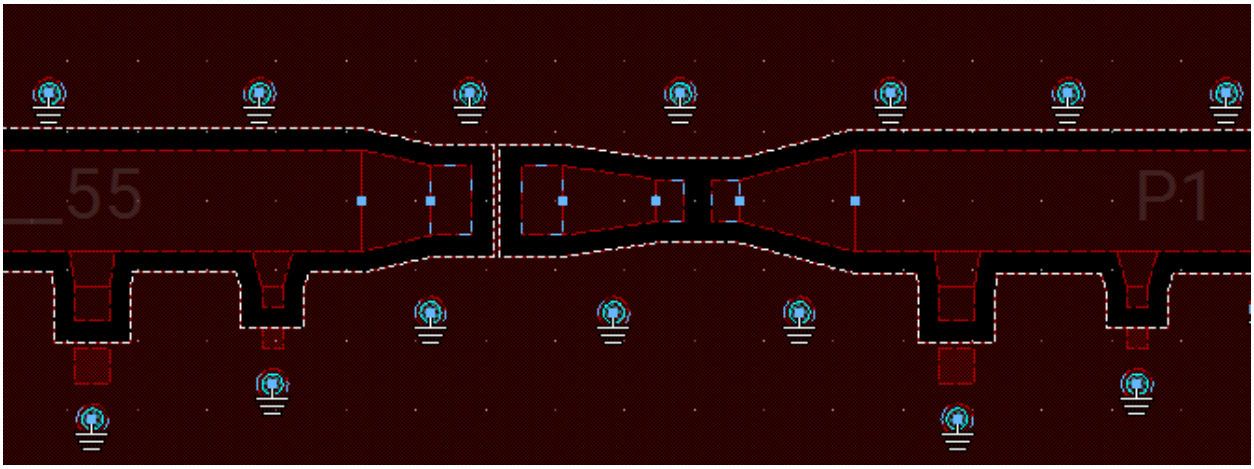


Figure 3.2.5.4 – L1 and L5 Lumped Pad Layout

3.2.6 Bias Tee

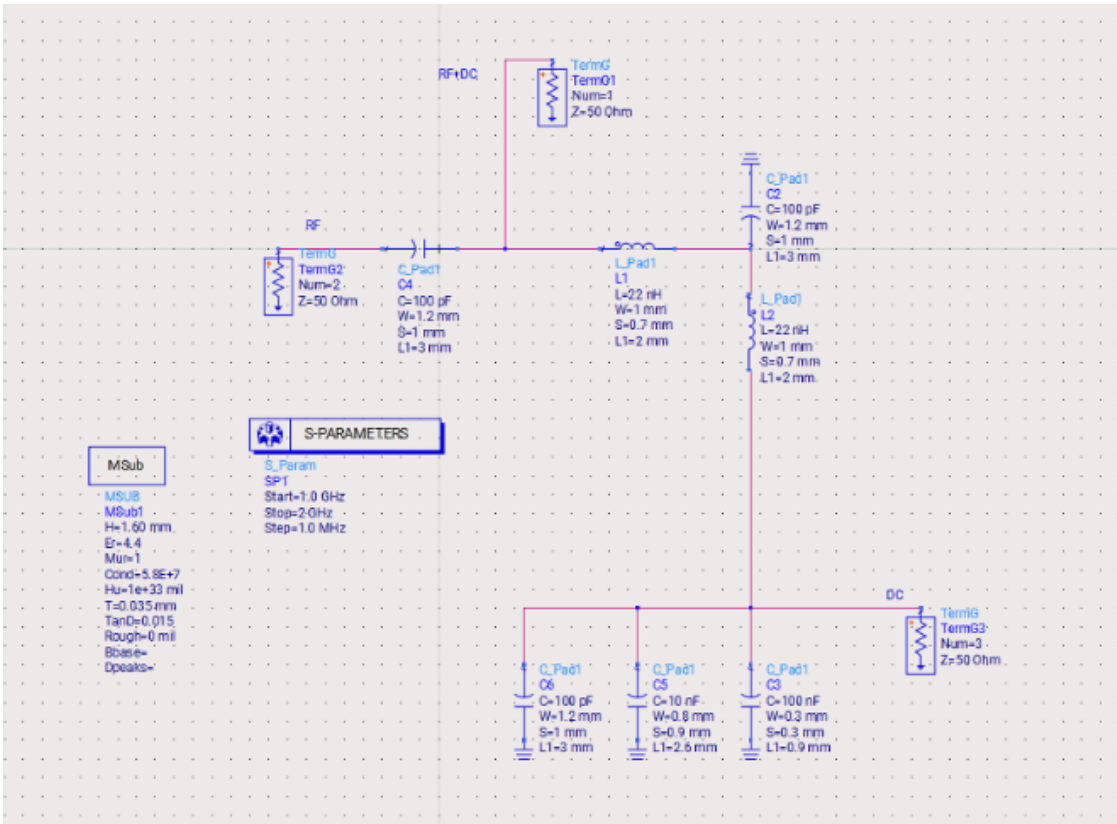


Figure 3.2.6.1 – Bias Tee Circuit Schematic ADS

The bias tee circuit is used to combine DC power with the RF signal on a single transmission line. This is achieved using a series capacitor in the RF path, which blocks DC but allows RF signals to pass, and an inductor in the DC branch, which allows DC current to flow while presenting a high impedance to RF signals. Additional shunt capacitors are included along the DC line to provide a wideband low impedance path to ground for our desired frequency, mid band frequency harmonics and low frequency noise. This configuration ensures efficient RF signal transmission while providing stable DC biasing to the LNA.

$Xc = 1/(2\pi f C)$ where C = 100pF (1Ω impedance to RF, much less than 50Ω).

$XL = 2\pi f L$ where L = 22nH (200Ω impedance to RF, much larger than 50Ω).

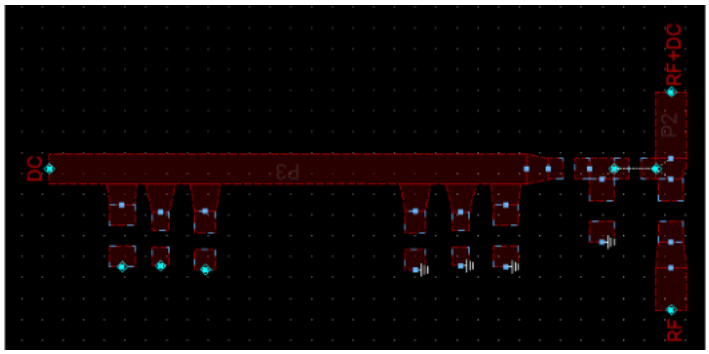


Figure 3.2.6.2 – Bias Tee Layout Design ADS

In the final PCB layout, the RF output and DC biasing point of the LNA are distanced relatively far apart, which can be problematic at high frequencies. To further improve RF stability, a multi-stage decoupling network consisting of 100nF, 10nF, and 100pF capacitors was implemented. These capacitors were placed both at the bias tee junction and again close to the LNA since the DC trace spans a relatively long distance. This ensures broadband filtering of noise across different frequency ranges and stabilizes the DC supply at the point of use.

4 Results and Validation

4.1 Prototype

The final prototype consists of three final PCBs designed and ordered through JLPCB, as well as four other modular PCBs that were planned to be used for testing. All boards were designed on a standard 2-layer FR-4 substrate with a thickness of 1.6mm except for L5 which was stacked to have a height of 3.2mm. The physical boards have not arrived due to manufacturing and international shipping issues. As a result, Gerber file renders and 3D board models have been presented in Figures 4.1.1 to Figure 4.1.8 to serve as the final prototype documentation.

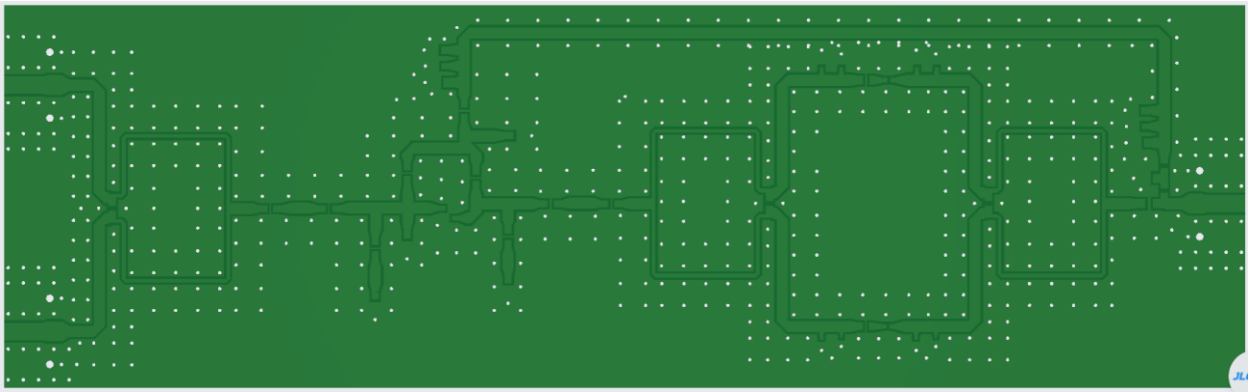


Figure 4.1.1– JLPCB Gerber View of PCB1 - RF Signal Processing Board with SAW Bandpass Filters

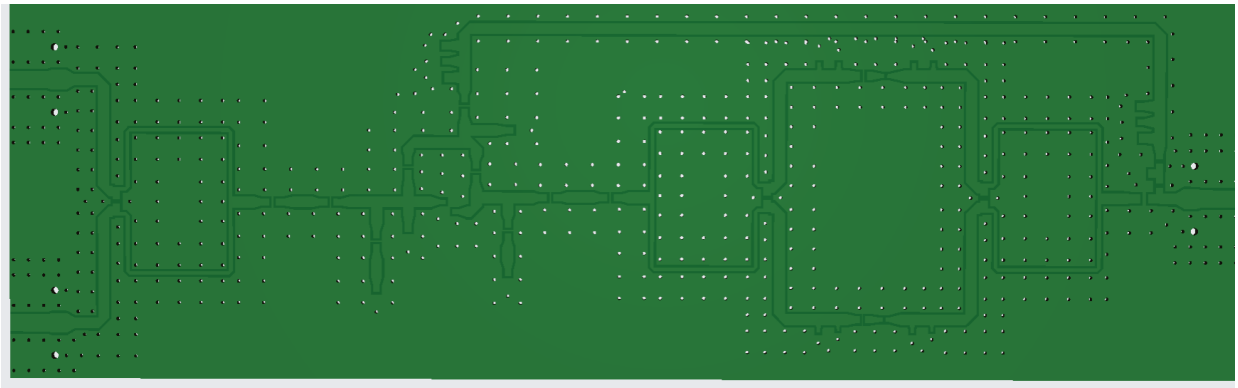


Figure 4.1.2– JLPCB Gerber View of PCB2 - RF Signal Processing Board with Lumped Bandpass Filters

Figure 4.1.3– PCB3 – Combined PCB with L1, L5 and RF Signal Processing Stages

Figure 4.1.4– JLPCB Gerber View of Dedicated L1 Patch Antenna PCB

Figure 4.1.5– JLPCB Gerber View of Dedicated L5 Patch Antenna PCB

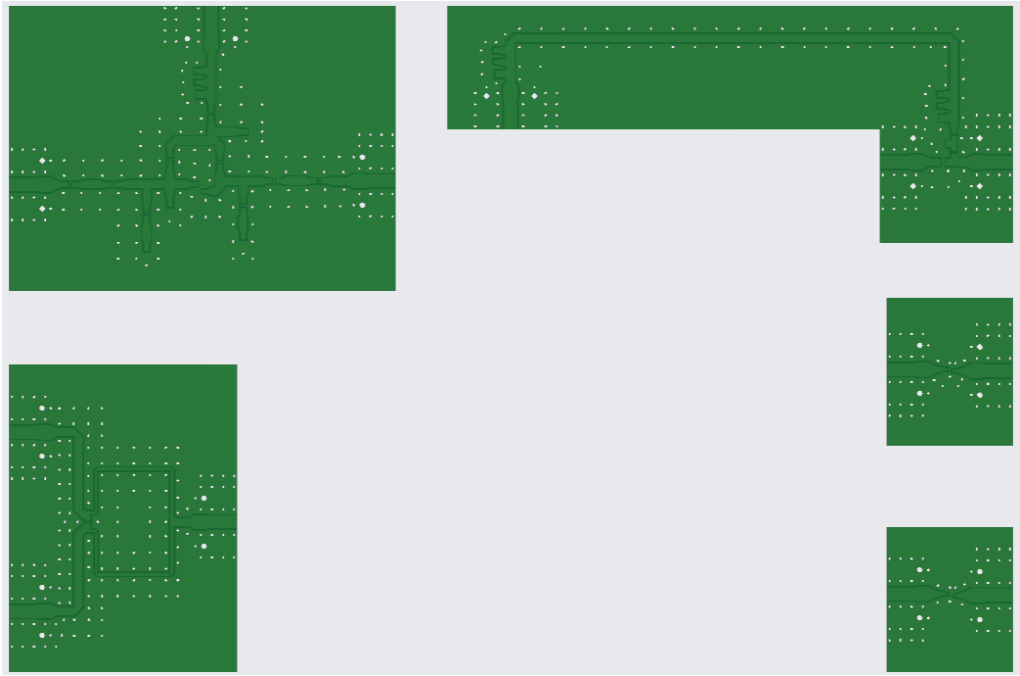


Figure 4.1.6– JLPCB Gerber View of Modular RF Signal Processing Board

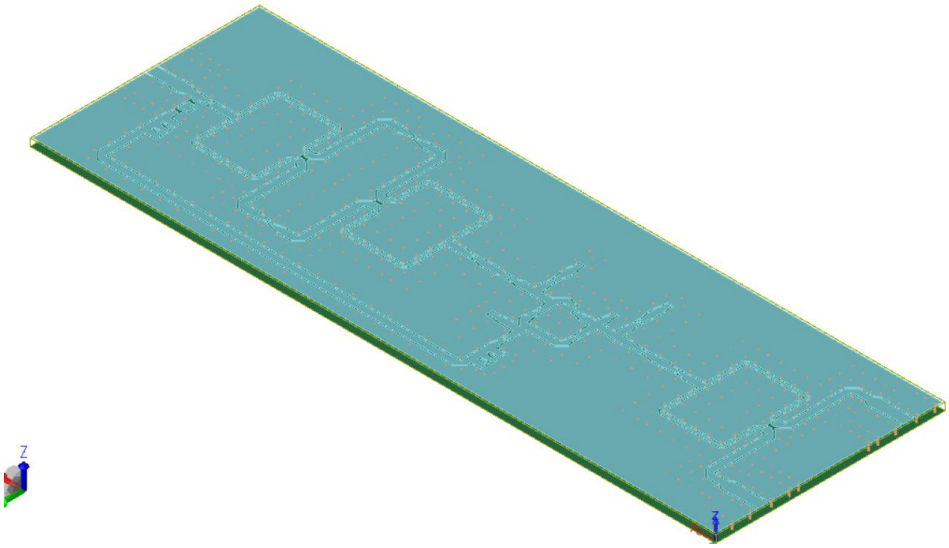


Figure 4.1.7– ADS – Final RF Signal Processing Board Front 3D View

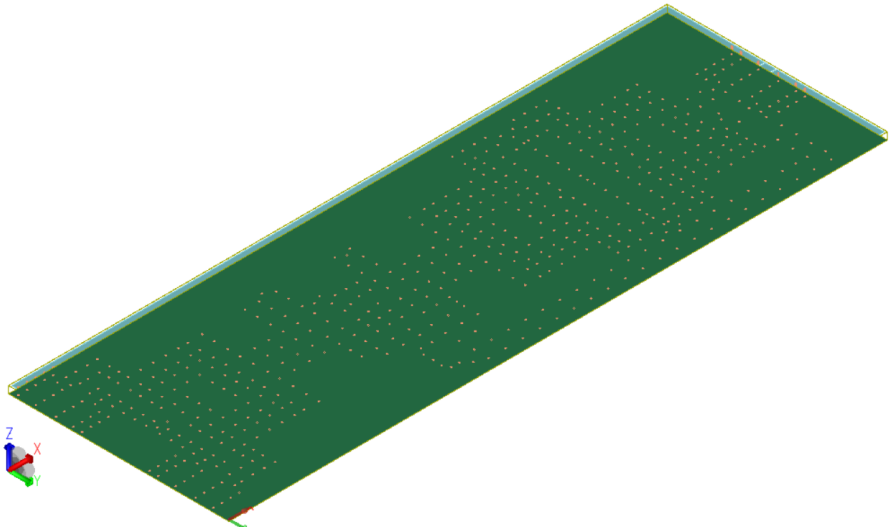


Figure 4.1.8– ADS – Final RF Signal Processing Board Back 3D View

Table 1: Component Cost Summary for RF Signal Processing Board (Digi Key & Mouser)

Component Description	**Quantity**	**Total Cost (CAD)**
4.7 kΩ resistor	10	1.62
2.2 kΩ resistor	10	1.62
10 kΩ resistor	10	1.62
5.6 kΩ resistor	5	1.30
8.2 kΩ resistor	3	0.78
3.9 kΩ resistor	3	0.45
270 Ω resistor	3	0.78
100 Ω resistor	20	1.62
10 Ω resistor	10	0.52
7.15 Ω resistor	3	0.45
5.1 Ω resistor	10	0.49
1.5 Ω resistor	3	0.54
10 µF capacitor	10	3.52
1 µF capacitor	10	0.90

0.1 µF capacitor	10	0.79
100 pF capacitor	20	0.40
24 pF capacitor	10	0.55
20 pF capacitor	3	0.51
12 pF capacitor	3	0.51
7 pF capacitor	10	0.92
5 pF capacitor	3	0.51
3 pF capacitor	3	0.51
2 pF capacitor	10	0.92
1.8 pF capacitor	3	0.51
1.6 pF capacitor	10	0.92
1.5 pF capacitor	10	0.92
1 pF capacitor	10	0.92
10000 pF capacitor	10	0.04
100 nH inductor	10	1.66
22 nH inductor	10	1.30
15 nH inductor	10	1.30
9.1 nH inductor	5	1.00
8.4 nH inductor	5	1.00
8.2 nH inductor	10	1.42
6.8 nH inductor	10	1.66
6.2 nH inductor	3	0.48
RF transistor	10	4.91
SMA edge-mount connector	10	45.71
SMA to SMA coaxial cable (6")	4	71.04
Signal Conditioning Signal Automotive SAW FILTER, GNSS L5/E5a/B2a	3	6.27
Signal Conditioning Automotive SAW FILTER, GNSS L1	3	6.27

Table 2: PCB Fabrication Cost Summary (JLPCB)

Component Description	**Quantity***	**Total Cost (CAD)**
PCB1 - RF Signal Processing Board with SAW Bandpass Filters (Figure 4.1.1)	5	7.10
PCB2 - RF Signal Processing Board with Lumped Bandpass Filters (Figure 4.1.2)	5	7.30
PCB3 – Combined PCB with L1, L5 and RF Signal Processing Stages (Figure 4.1.3)	5	21.30
Dedicated L1 Patch Antenna PCB (Figure 4.1.4)	5	2.00
Dedicated L5 Patch Antenna PCB Top (Figure 4.1.5)	5	4.00
Dedicated L5 Patch Antenna PCB Bottom (Figure 4.1.5)	5	4.00
Modular RF Signal Processing Board (Figure 4.1.6)	5	13.10

**The minimum printing quantity on JLCPCB is 5*

4.2 Validation/Testing Strategy or Protocols

The validation strategy focuses on experimentally verifying antenna performance and filtering functionality, as these components present the highest sensitivity to fabrication tolerances and high-frequency parasitic effects. Testing will be conducted iteratively, allowing design refinements between fabrication cycles if required.

For the patch antennas, validation begins with physical inspection. Fabricated dimensions (length and width) will be measured using digital calipers and compared directly to the final ADS design values to ensure manufacturing accuracy. Since antenna resonance is highly dependent on geometry, even small deviations can shift the operating frequency.

If the resonance frequency is correct but impedance matching is poor (i.e., insufficient return loss), the inset feed position will be adjusted in future iterations. Specifically, the inset

depth will be varied in small increments (≈0.2 mm) to better locate the true 50 Ω feed point. In the worst-case scenario, multiple revised antenna designs will be fabricated using a "bracketed" approach, where patch dimensions are varied by approximately ±1% to account for fabrication tolerances and material uncertainties. This ensures convergence to the target frequency and impedance match.

For the bandpass filters, validation follows a two-step approach due to practical limitations encountered at GHz frequencies. Initial testing will be performed on a realizable lumped-element bandpass filter design, acknowledging that performance may be degraded compared to ideal simulations due to parasitic inductance and capacitance. S-parameter measurements (primarily S21 and S11) will be used to evaluate insertion loss, bandwidth, and out-of-band rejection.

Given the limitations of both lumped and transmission-line implementations (e.g., impractically small component values and trace widths), commercial SAW filters will be used as a backup solution. These filters, sourced from manufacturers such as Qorvo or Digi-Key, offer more predictable high-frequency performance. A dedicated PCB layout including appropriate pad sizes will allow direct integration and testing of these components.

Performance comparison between the lumped filters and SAW filters will be conducted using identical measurement setups. This will determine whether the custom filter design meets system requirements or if the SAW filters provide a superior and more reliable alternative.

4.3 Final Test Results

The manufacturing of the custom PCBs was significantly delayed due to repeated automated holds on the JLCPCB platform, which often contradicted manual assurances provided by customer support via email. As of the time of writing, four of the seven boards remain on hold, while the three completed boards are stored in a warehouse pending release of the remaining units.

Although JLCPCB representatives confirmed on multiple occasions, specifically on March 20th, March 23rd, and April 2nd, that the order would proceed "ASAP" following clarification of intentional design choices (such as the absence of drill holes and certain dimensional discrepancies), the order status continued to display as "suspended" or "on hold" for over a week.

jenny@jlcpcb.com Fri, Mar 20 00:40 (5 days ago)

Hi Yassine， Thanks for your confirmation. We will proceed with this order ASAP. Have a nice day. Best regards, Jenny On Fri, …

Yassine _ <yassinebenyassine2003@gmail.com> Mon, Mar 23, 8:13 PM (2 days ago)

À jenny

Hi Jenny,

I just logged into the website and noticed that my order has been suspended this entire time without any email or notification. This is quite concerning, especially since I paid for express delivery.

I'm very disappointed with the lack of communication, as I have an engineering capstone showcase this Thursday, and this delay is causing significant issues for me.

Could you please look into this urgently and provide an update on the status of my order?

Figure 4.3.1 - Communication with JLCPCB

serena@jlcpcb.com lun. 23 mars 23:42

À moi

Dear Customer,

Thanks for your kind reply.

Well noted. We will proceed with this order per your request ASAP.

Thanks for your kind support and have a nice day💗 😊

Best Regards,

Serena

Figure 4.3.2 - Communication with JLCPCB

edward@jlcpcb.com ven. 3 avr. 02:38 (il y a 6 jours)

À moi

Dear Yassine,

Thank you for your message. The system automatically cancels orders if any issues are detected in the uploaded Gerber files to prevent production errors.

I can confirm that your original files have been successfully re-uploaded, and I have notified our engineering team. Your order is now ready to proceed without any further issues.

Best regards,

Edward TEE

Figure 4.3.3 - Communication with JLCPCB

This issue appears to stem from the manufacturer's automated screening system, which flags designs with missing layers (e.g., solder mask) to prevent potential production errors. Despite explicitly requesting that the boards be fabricated as submitted due to strict capstone project deadlines, the mismatch between manual approval and automated validation resulted in prolonged delays.

As a result, the team was unable to obtain a functional prototype in time for the capstone showcase and shifted focus toward securing at least a partial prototype for the final report.
As a last resort, fabrication was attempted through the Western Engineering Electronics Shop, despite known limitations in achieving the required RF precision. Following consultation, it was determined that only the patch antenna could be produced, as the coplanar waveguide (CPWG) clearances on the signal processing board were below the shop's minimum requirement of 20 mils.

However, the fabricated boards were ultimately unusable. The milling process was unable to remove sufficient copper without risking damage to the equipment, resulting in boards that remained largely unetched (see Figure 4.3.4). Robert from the electronics shop concluded that the facility did not have the capability to manufacture the design within the required specifications. Unless we were to manually scrape the copper with a knife for hours, we are stuck with useless boards.

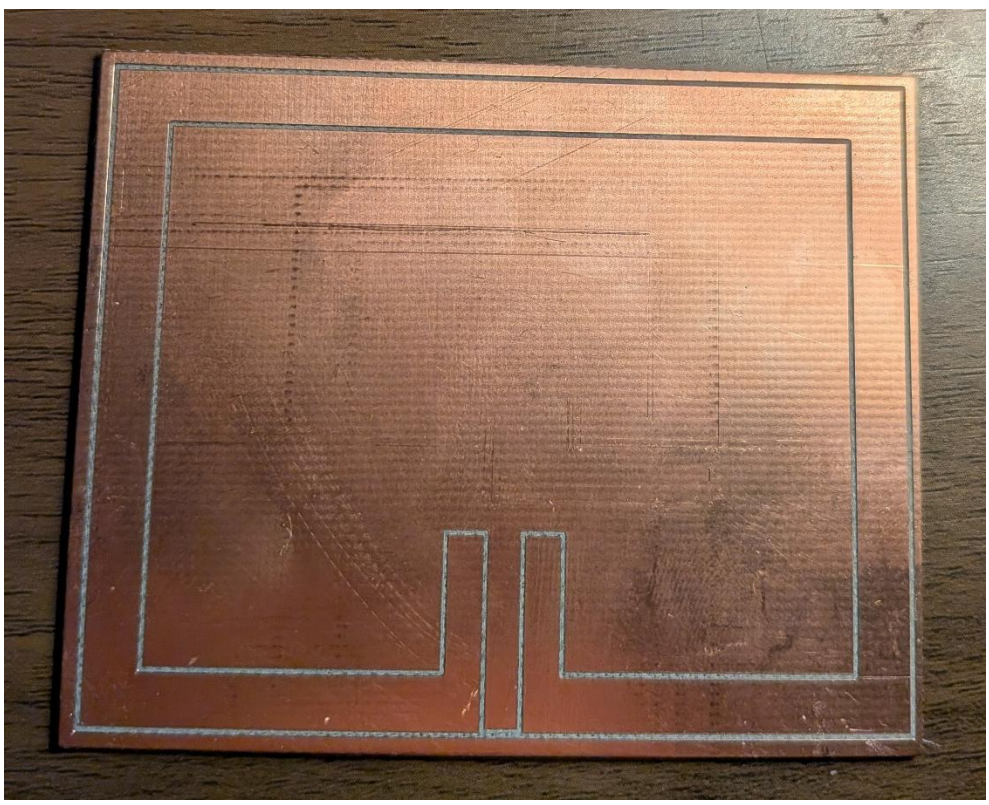


Figure 4.3.4 – L1 board manufactured by the Electronics Shop

As such, only simulation results are presented in the following section.

4.3.1 L1 and L5 Patch Antenna

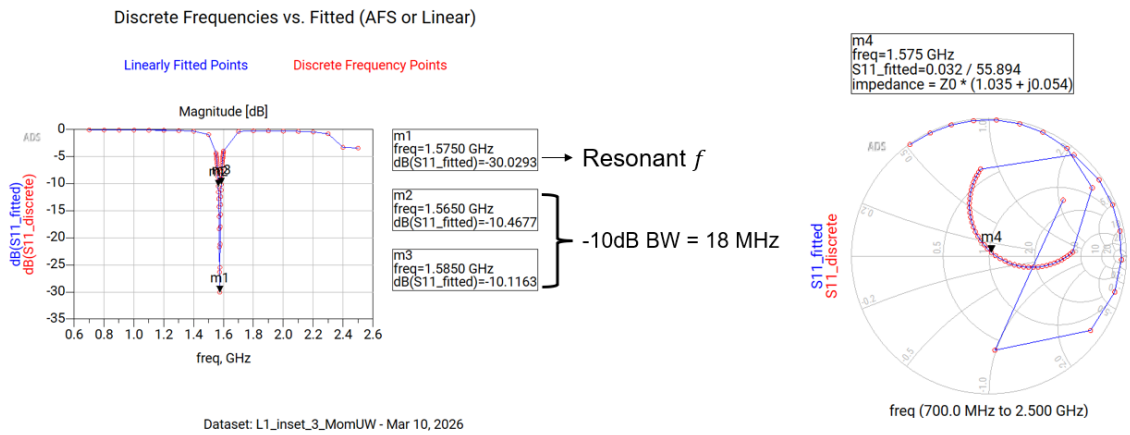
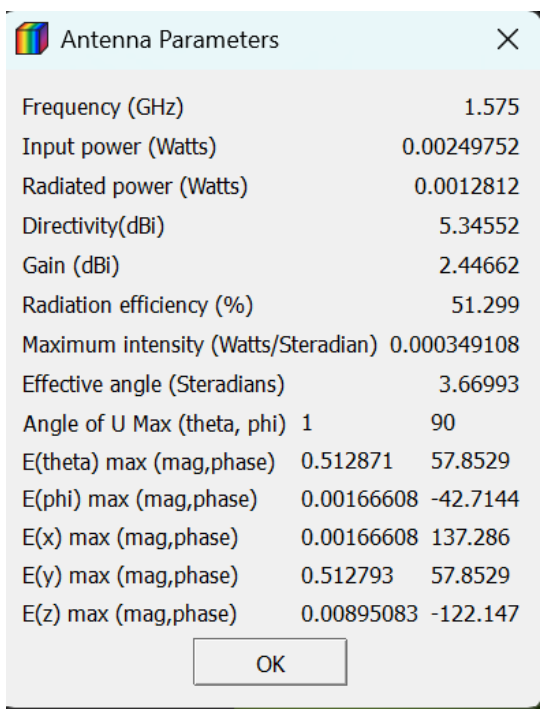


Figure 4.3.1.1 – L1 Return Loss Simulation Results



Antenna Parameters

Frequency (GHz)		1.575
Input power (Watts)		0.00249752
Radiated power (Watts)		0.0012812
Directivity(dBi)		5.34552
Gain (dBi)		2.44662
Radiation efficiency (%)		51.299
Maximum intensity (Watts/Steradian)		0.000349108
Effective angle (Steradians)		3.66993
Angle of U Max (theta, phi)	1	90
E(theta) max (mag,phase)	0.512871	57.8529
E(phi) max (mag,phase)	0.00166608	-42.7144
E(x) max (mag,phase)	0.00166608	137.286
E(y) max (mag,phase)	0.512793	57.8529
E(z) max (mag,phase)	0.00895083	-122.147

OK

Figure 4.3.1.2 – L1 Antenna Parameters

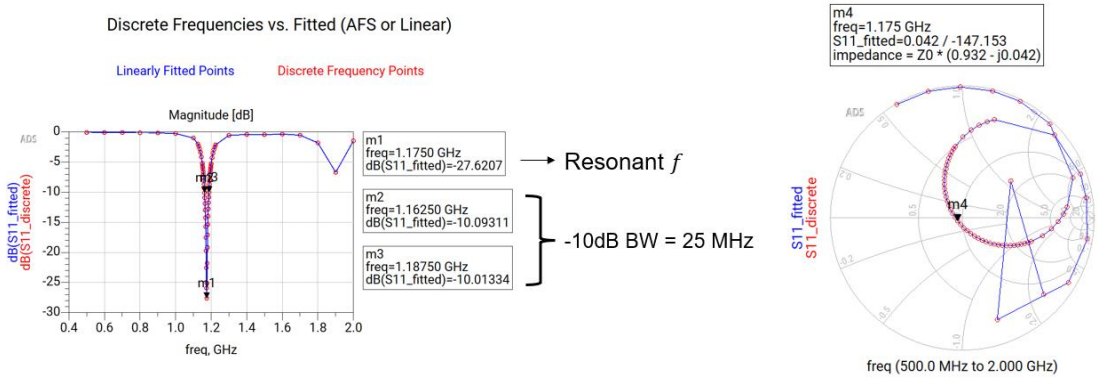
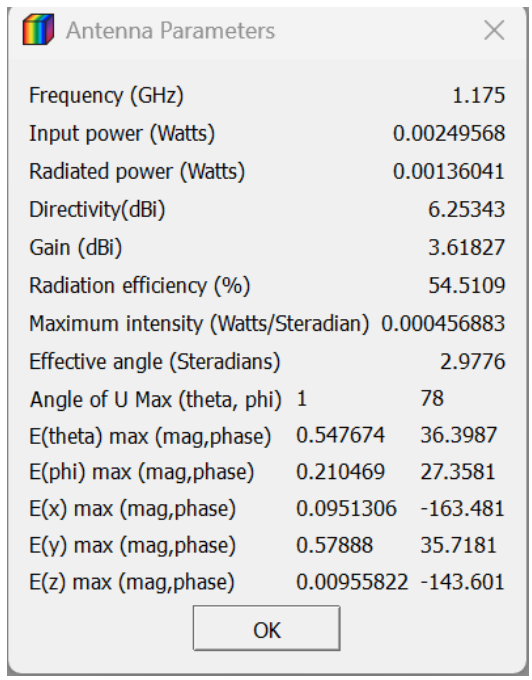


Figure 4.3.1.3 – L5 Return Loss Simulation Results



Antenna Parameters

Frequency (GHz)		1.175
Input power (Watts)		0.00249568
Radiated power (Watts)		0.00136041
Directivity(dBi)		6.25343
Gain (dBi)		3.61827
Radiation efficiency (%)		54.5109
Maximum intensity (Watts/Steradian)		0.000456883
Effective angle (Steradians)		2.9776
Angle of U Max (theta, phi)	1	78
E(theta) max (mag,phase)	0.547674	36.3987
E(phi) max (mag,phase)	0.210469	27.3581
E(x) max (mag,phase)	0.0951306	-163.481
E(y) max (mag,phase)	0.57888	35.7181
E(z) max (mag,phase)	0.00955822	-143.601

OK

Figure 4.3.1.4 – L5 Antenna Parameters

The simulated L1 and L5 antennas achieve very low return losses of -30 dB and -27.6 dB respectively which means that less than 0.1% and 0.17% of the respective band signal gets reflected at the input, indicating excellent impedance matching. Roughly half of the accepted power is successfully radiated, which is typical for compact PCB antennas and confirms that losses are present but not excessive.

The gain values in the ~2.5–3.6 dBi range indicate useful but not highly directive radiation, which aligns well with GNSS requirements where wide sky visibility is more important than high directivity. This is further supported by the relatively broad radiation

patterns, ensuring the antenna can receive signals from multiple satellite angles rather than favoring a narrow direction.

The difference between directivity and gain suggests that efficiency is the main limitation, meaning the antenna design itself is effective, but material losses could still be improved. After all, almost all commercial patch antennas are made of ceramic, not epoxy glass.

The biggest area of improvement of the antennas remains their linear polarization, as GNSS requires RHCP, thus resulting in 3 dB loss due to polarization mismatch.

4.3.2 Wilkinson Power Divider and Combiner

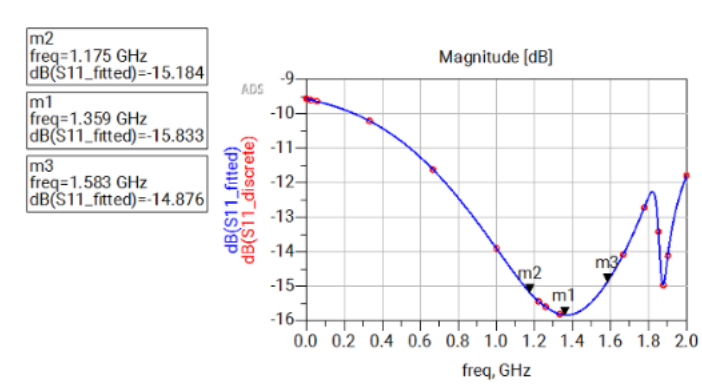


Figure 4.3.2.1 – Wilkinson S11

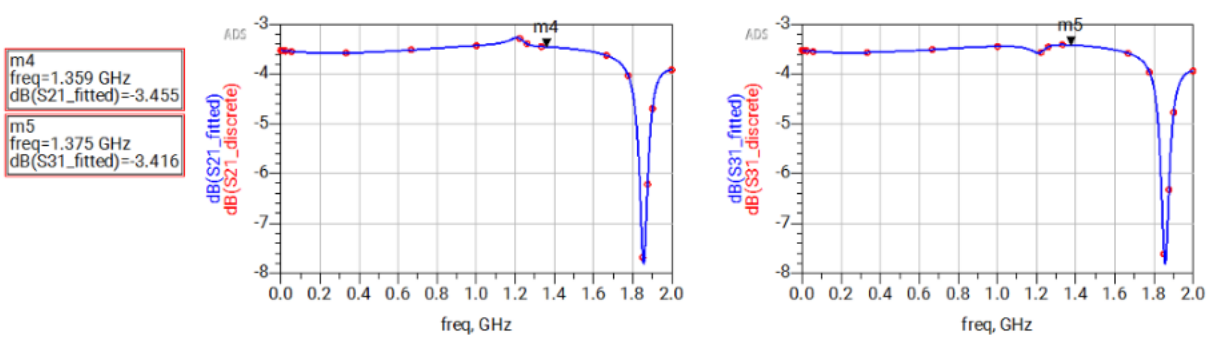


Figure 4.3.2.2 – Wilkinson S21 and S31

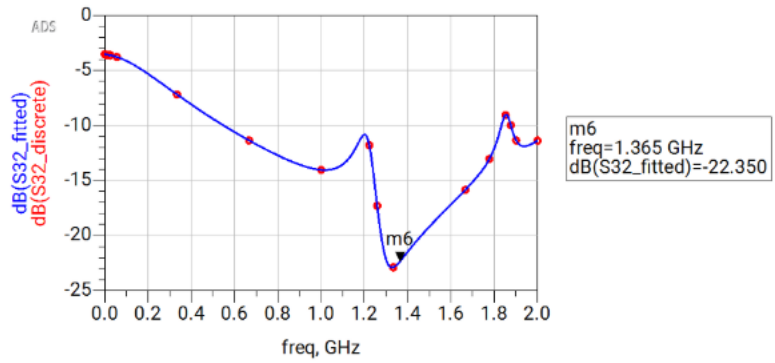


Figure 4.3.2.3– Wilkinson S32

Preliminary performance of the Wilkinson power divider was evaluated using ADS Momentum microwave simulation based on the final CPWG layout. The input return loss S11 in Figure 4.3.2.1 shows good matching across the operating region, with values of approximately -15 dB across L1 and L5. These values are good because they indicate that only a small portion of the input power is reflected to the source, meaning the system is well matched to 50 ohms. Since all values are below -10 dB, the matching is acceptable for the intended GNSS application.

The transmission coefficients S21 and S31 are approximately -3.4 dB GHz, respectively. This is close to the ideal -3 dB expected from an equal split Wilkinson divider, where the input power is divided evenly between the two output ports. This additional loss is expected in a practical CPWG layout due to non ideal losses in the physical structure. As well as due to some of the non ideal geometry causing parasitic losses like at the main junction with the isolation resistor. The fact that S21 and S31 are almost identical also confirms that the divider is providing a balanced split between the two branches.

The isolation parameter S32 between output ports for L1 and L5 reaches –22 dB at 1.36GHz, which indicates strong isolation between the two output ports. This is an important result because one of the key advantages of a Wilkinson divider is that signals appearing at one output should not significantly couple with the other output. Good isolation improves port independence and helps preserve signal integrity in the overall RF chain.

Overall, these Momentum simulation results indicate that the Wilkinson divider meets the expected design behavior with good input matching, near equal power split, and strong output isolation near the design frequency. This suggests the layout is suitable for fabrication and further validates the final prototype.

4.3.3 Wideband LNA

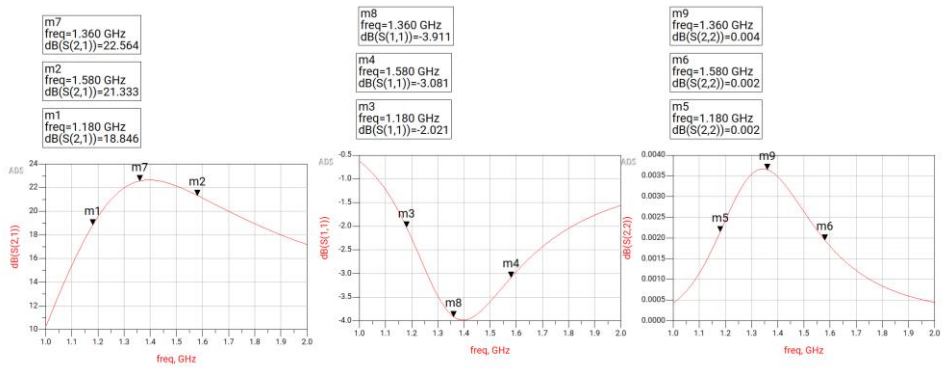


Figure 4.3.3.1 – LNA Schematic Simulation Results

The simulation results demonstrate that the designed LNA achieves strong performance across the desired GNSS frequency bands. The forward gain S21 remains high, reaching approximately 22.6 dB at 1.36 GHz and maintaining over 18 dB at 1.18 GHz and over 21 dB at 1.58 GHz, indicating effective amplification across both the L5 and L1 bands. The input reflection coefficient S11 is below −2 dB across the operating range, with a minimum near −4 dB, showing reasonable impedance matching at the input. Additionally, the output reflection coefficient S22 remains close to 0 dB, indicating stable output behavior with minimal variation. Overall, these results confirm that the LNA provides sufficient gain and acceptable matching over the required frequency range, making it suitable for wideband GNSS applications.

4.3.4 L1 and L5 Bandpass Filters

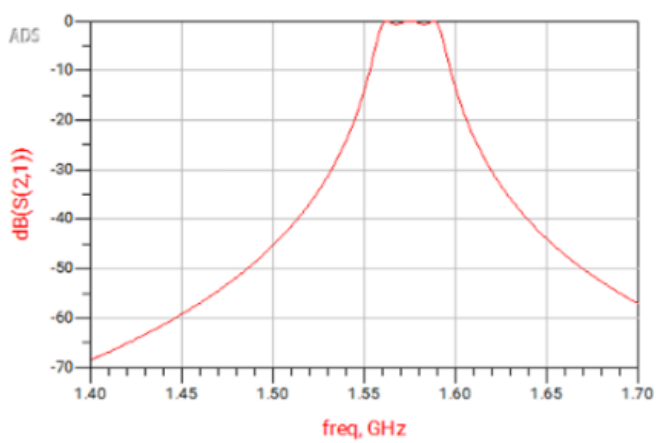


Figure 4.3.4.1– L1 Lumped Bandpass Filter

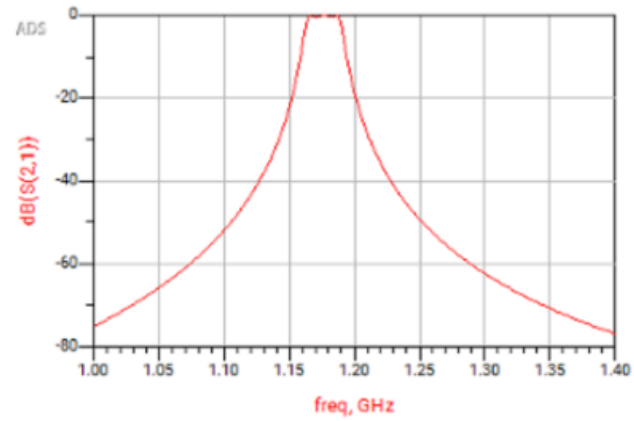


Figure 4.3.4.2– L5 Lumped Bandpass Fitler

The L1 and L5 bandpass filters were designed and simulated in ADS using lumped element models to target the GNSS frequencies of 1.575 GHz (L1) and 1.176 GHz (L5). The S21 responses show clear bandpass behavior, with low insertion loss near the center frequency and strong attenuation outside the passband. This demonstrates effective rejection of out-of-band signals, which is critical for reducing interference and isolating bands.

However, it is important to note that at these frequencies, lumped components as small as these are highly affected by parasitic and layout effects, which are not fully captured in ideal simulations.

To mitigate this risk and validate system performance, commercial SAW filters were also purchased as a backup solution. Refer to Appendix Section 9.1 for SAW filter specifications and results. These components offer more predictable high frequency performance compared to discrete lumped implementations. PCBs were made using both the LC bandpass filters and purchased SAW, and results will be compared.

4.3.5 Bias Tee

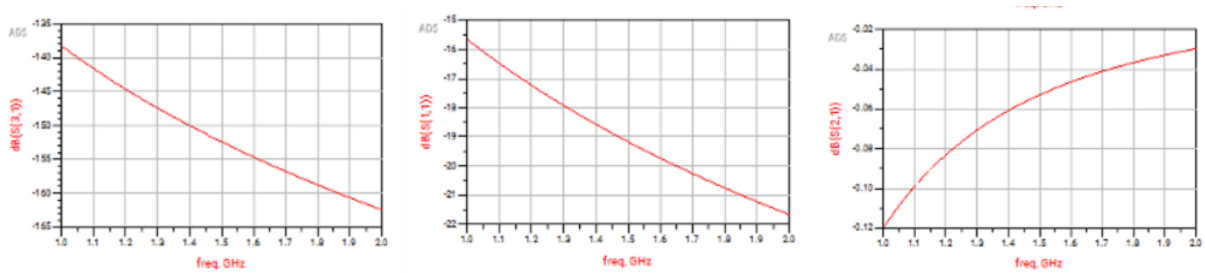


Figure 4.3.5.1– Bias Tee Simulation Results

The bias tee was simulated in ADS using S-parameter analysis to verify proper RF/DC separation. The results confirm that the circuit successfully injects DC while minimally disturbing the RF signal path. The input return loss S11 is below -15 dB across L1 and L5, indicating solid impedance matching to 50ohms. The RF transmission S21 remains close to 0 dB across the frequency range. This is a strong result, as it indicates that the RF signal passes through the bias tee with negligible attenuation, which is critical for maintaining signal strength. The isolation to the DC port S31 is very high (-140dB) demonstrating strong suppression of RF leakage into the DC supply. This confirms that the inductors effectively block RF signals while allowing DC biasing of the LNA.

Overall, the simulated results indicate that the bias tee performs as intended, providing low loss RF transmission, strong RF/DC isolation, and stable DC biasing, making it suitable for integration in the RF front end.

4.3.6 Overall System Results

To validate the complete RF signal processing chain, the S-parameter data from each individual simulated system stage was exported as S2p files and cascaded together with a full schematic in ADS as shown in Figure 4.3.6.1. It should be noted that this is a circuit level simulation using exported S-parameter blocks rather than a full EM simulation and therefore does not capture all non-ideal effects that may be present in the physical PCB. However, the results in Figure 4.3.6.2 demonstrate clear dual passband behavior with desired amplification in L1 and L5 frequency bands, as well as strong out of band rejection.

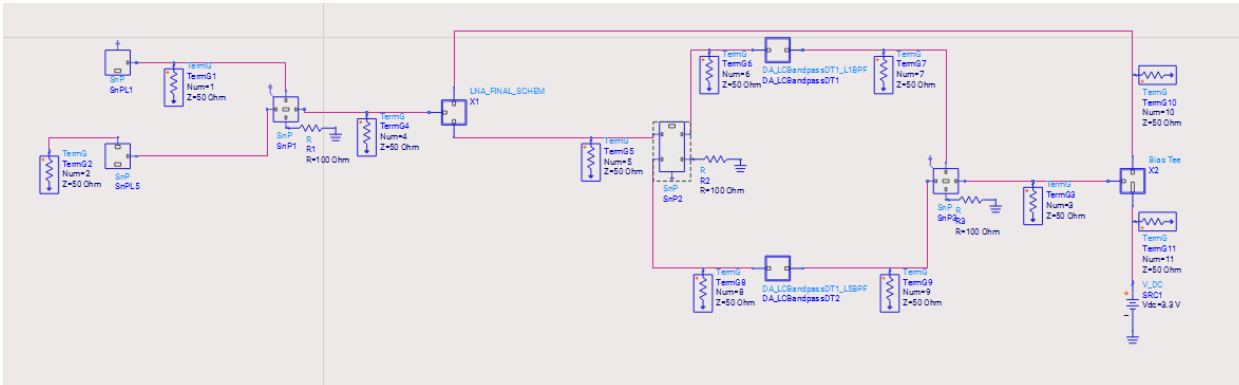


Figure 4.3.6.1 – ADS Final System Schematic with Exported Simulated Results

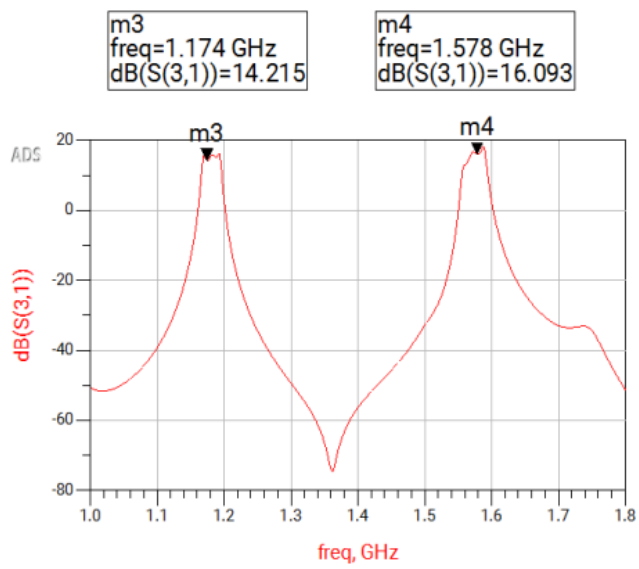


Figure 4.3.6.2– ADS Final System Schematic with Exported Simulated Results

5 Project Team Management

5.1 Team Member Task Breakdown and contributions and timelines

Task allocation for the project was initially established during the proposal and midterm stages, then refined as the work progressed from concept development into detailed RF analysis, subsystem integration, PCB layout, fabrication planning, and final validation preparation. In the earlier stages of the project, responsibilities were broader and focused primarily on feasibility analysis, antenna research, circuit simulation, and system-level planning. As the final architecture became fixed, each team member's role became more closely aligned with a specific

part of the RF front-end, while report preparation, design reviews, and major technical decisions remained shared across the group.

Atrin Shahrokh contributed primarily to the early-stage development of the antenna subsystem and the overall system architecture. This included research into GNSS antenna types, placement considerations, and design constraints associated with dual-band L1/L5 operation in an automotive environment. Atrin also supported the setup of the FEKO and ADS modelling workflow, contributed to block-diagram development and subsystem organization, and assisted with LNA-related component sizing and footprint standardization to support PCB implementation.

Yassine Benjelloun was primarily responsible for the antenna design and simulation work. His contributions included selection of the PCB stack-up based on supplier capabilities and electrical requirements, calculation of initial patch dimensions, iterative tuning of the L1 and L5 microstrip patch antennas in ADS, and verification of impedance-matching performance through simulation. As the project advanced, this role remained focused on ensuring that the antenna subsystem met the required resonant frequencies and could be integrated with the RF signal-processing board.

Brandon Medeiros was primarily responsible for the RF signal-processing portion of the design. This included the design and simulation of the dual-band bandpass filters, Wilkinson combiner and divider stages, bias tee, and overall signal path between the antenna stage and receiver interface. Brandon also completed PCB substrate and transmission-line calculations, contributed to full-system ADS simulation, and carried out layout development and EM simulation for the RF signal-processing board, including implementation of grounded coplanar waveguide traces and practical layout refinements required for fabrication.

Jordan Austin focused primarily on system integration, validation planning, and prototype implementation support. Early contributions included development of system-level validation considerations and support for defining the combined antenna–LNA simulation objectives. In the later stages of the project, this role expanded to include coordination of subsystem integration, preparation of measurement and testing procedures, and compilation of the bill of materials and

component requirements for prototype assembly. These contributions supported the transition from simulation-driven design to a manufacturable and testable hardware implementation.

In addition to these primary responsibilities, several tasks were completed collaboratively across the team. These included the literature review, concept evaluation, simulation review, technical decision-making, preparation of design review material, report writing, and planning for prototype fabrication and validation. By the final stage of the project, the team had completed the design of the L1 and L5 antenna boards, finalized the main RF front-end layouts, and ordered the required PCBs for fabrication, with hardware validation planned as the final step following successful simulation results across all major subsystems.

5.2 Team Communication and Organization Methodology

We used a combination of synchronous and asynchronous communication to achieve efficient communication and organization throughout the project. For day-to-day communication, Discord and Instagram were used for quick updates, informal discussions, task coordination, and meeting planning. Discord was also used for quickly sharing files and facilitating online group meetings when in-person meetings were not possible.

Email was used for formal communication, particularly when interacting with our faculty advisors, Dr. Ali Attaran and Dr. Ken McIsaac, Ph.D., P.Eng., including scheduling meetings and providing project updates. Microsoft Teams was also utilized for meetings when an advisor(s) could not meet in person to present simulations, schematics, and technical results.

5.3 Meetings/Attendance Listing

Provide a list/table of meetings and attendance (as a table), including meetings with the faculty advisors. Meeting confirmation (emails) with faculty advisors can be listed in the appendix.

Date	Meeting type	Attendees	Key Outcomes
Sep 19, 2025	Group Meeting (in-person)	AS, YB, JA, BM	Identified GNSS antenna project; contacted and secured Dr. Attaran as faculty advisor; defined initial project scope
Oct 6, 2025	Advisor meeting	AS, YB,	Presented project concept and antenna/LNA

	(in-person)	JA, BM	approach; received approval and design refinement guidance
Oct 21, 2025	Group Meeting (in-person)	AS, YB, JA, BM	Discussed antenna solutions; delegated tasks for project proposal completion; scheduled advisor meeting
Oct 27, 2025	Advisor meeting (in-person)	AS, YB, BM	Reviewed antenna simulations; corrected ADS setup; introduced FEKO; discussed filtering and next design steps
Nov 11, 2025	Group Meeting (in-person)	AS, JA, BM	Finalized antenna simulation targets; planned combined LNA-antenna simulations
Nov 18, 2025	Group Meeting (in-person)	AS, YB, JA, BM	Discussed filter designs and simulation; worked together completing design review slides & presentation
Dec 1, 2025	Group Meeting (online)	AS, YB, JA, BM	Reviewed circuit simulation results and discussed possible design refinements
Dec 7, 2025	Advisor meeting (online)	AS, YB, JA, BM	Updated Dr. Attaran on our progress; discussed next steps
Jan 20, 2026	Group Meeting (in-person)	AS, YB, JA, BM	Reviewed progress from first term and confirmed the final hybrid dual-band architecture. Reassigned second-term tasks for antenna tuning, LNA layout, filtering, and validation planning.
Jan 26, 2026	Group Meeting (in-person)	AS, BM	Worked on the wideband LNA PCB layout, including component footprints, placement, and routing considerations for manufacturability.
Jan 26, 2026	Group Meeting (online)	AS, BM, YB	Reviewed antenna simulation progress and discussed integration of the L1 and L5 antenna boards with the RF signal-processing board.

Jan 29, 2026	Group Meeting (online)	AS, YB, BM	Discussed remaining antenna tuning and impedance-matching adjustments. Reviewed whether further layout changes were needed before moving closer to fabrication.
Feb 1, 2026	Group Meeting (online)	AS, JA, BM	Reviewed subsystem interconnection strategy and discussed validation planning for the combined antenna, LNA, and filter stages.
Feb 5, 2026	Group Meeting (online)	AS, YB, JA, BM	Compared subsystem simulation results and discussed practical filtering options, including backup SAW filters for improved high-frequency implementation reliability.
Feb 11, 2026	Group Meeting (in-person)	AS, YB, JA, BM	Reviewed RF board layout progress, including GCPW trace dimensions, Wilkinson sections, and grounding/via-stitching strategy.
Feb 18, 2026	Group Meeting (online)	AS, YB, JA, BM	Discussed prototype planning, receiver interface requirements, and documentation of design constraints for the final report.
Feb 27, 2026	Group Meeting (online)	AS, JA, YB	Reviewed final simulation outputs for major subsystems and discussed remaining issues to resolve before Design Review II and PCB ordering.
Mar 3, 2026	Group Meeting (in-person)	AS, YB, JA, BM	Reviewed final patch antenna tuning results and RF signal-processing board progress. Discussed layout refinements and preparation for fabrication.
Mar 13, 2026	Group Meeting (in-person)	AS, YB, JA, BM	Finalized material for Design Review II, reviewed subsystem results, and assigned remaining presentation and documentation

			tasks.
Mar 16, 2026	Advisor Meeting (in-person)	AS, YB, JA, BM	Presented project progress to Dr Attaran and Dr McIsaac
Mar 24, 2026	Group Meeting (online)	AS, JA, BM	Reviewed fabrication status and discussed how to present simulation-based validation results and design limitations in the final report.

6 Final Conclusions

The original objective was straightforward on paper: design a compact dual-band GNSS front-end operating at L1 and L5 that could realistically be built, tested, and eventually integrated into a vehicle. Whether the final outcome actually meets that bar depends on how strictly you interpret "meets."

In simulation, the answer is yes. Every subsystem cleared its target. The L1 and L5 patch antennas resonated at their respective frequencies with return losses of −30 dB and −27.6 dB, well below the −10 dB threshold required for acceptable impedance matching. The Wilkinson combiner produced a near-equal −3.4 dB power split across both bands with −22 dB port isolation at the design frequency. The wideband LNA delivered over 18 dB of gain at L5 and over 21 dB at L1, within the 20–30 dB target range. Both bandpass filters showed clear passband behavior centered at the correct frequencies with 30 dB or greater out-of-band rejection. The bias tee passed RF with negligible attenuation while maintaining −140 dB isolation from the DC supply. Cascaded together, the system showed dual-passband amplification at L1 and L5 with strong rejection everywhere else.

In hardware, the answer is incomplete. The PCBs were designed, reviewed, and ordered. They have not arrived. A last-resort attempt at in-house fabrication through the Western Engineering Electronics Shop produced boards that were unusable due to copper milling limitations. That outcome was frustrating but not surprising given the 0.3 mm clearances required by the GCPW layout. The shop was not equipped for it.

The gap between simulation and measured results is real. Lumped element filters at GHz frequencies will behave differently than ideal ADS models predict. The LNA S11 was below −2

dB rather than the −10 dB typically associated with well-matched RF inputs, and whether that translates to acceptable noise performance in a physical build is still an open question. The antennas are linearly polarized, which costs approximately 3 dB against RHCP satellite signals. That loss was a deliberate trade off against the complexity of implementing a hybrid coupler-based circular polarization scheme, but it is still a real system penalty.

What the project did produce is a fully designed, fabrication-ready dual-band GNSS front-end with documented simulation results across every stage. The architecture is modular enough to test subsystems independently once the boards arrive, and the SAW filter variants provide a reliable fallback if the lumped filters underperform. The design meets its simulation targets. Whether it meets its hardware targets is the one question this project ran out of time to answer.

7 Future Work and Recommendations

Upon arrival of the fabricated PCBs, the immediate next step is modular testing of each system stage using a vector network analyzer (VNA) to identify any discrepancies between simulated and measured S-parameters. Once each stage has been verified, full system testing of the assembled prototype can be completed to validate design targets. Following successful testing, the system will be evaluated in a real vehicle environment to assess positioning accuracy, signal quality and overall performance. As part of this testing phase, the antenna system will be mounted in various locations within a vehicle to determine which placements best balance RF signal integrity with aesthetics and stealth requirements.

Beyond the current prototype, several design improvements have been identified for future iterations. The RF signal processing board can be made far more compact with a stacked patch antenna opposed to having dedicated L1 and L5 antennas. Additionally, a pre LNA filtering stage can be introduced ahead of amplification to reduce noise configuration from strong interfering signals. This filtering stage is essential for testing the current antenna system with other nearby signals like LTE, FM/AM and other automotive RM before they reach the amplifier to provide stronger rejection. Finally, the long-term vision for the system would be to fully integrate custom antennas directly into the vehicle structure to enable a truly hidden and aesthetic design unlike the current skark fin antenna structure.

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9 Appendices

9.1 SAW Filter Specs

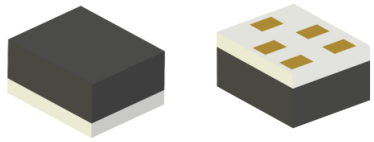

Figure 9.1.1 - SMD 1176.45 MHZ SAW Filter Product Image [24]

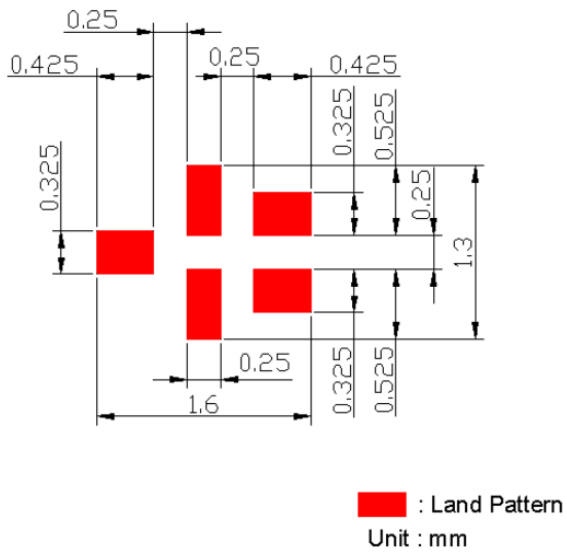

Figure 9.1.2 – L5 SAW Filter Land Pattern [24]

S21 narrow band

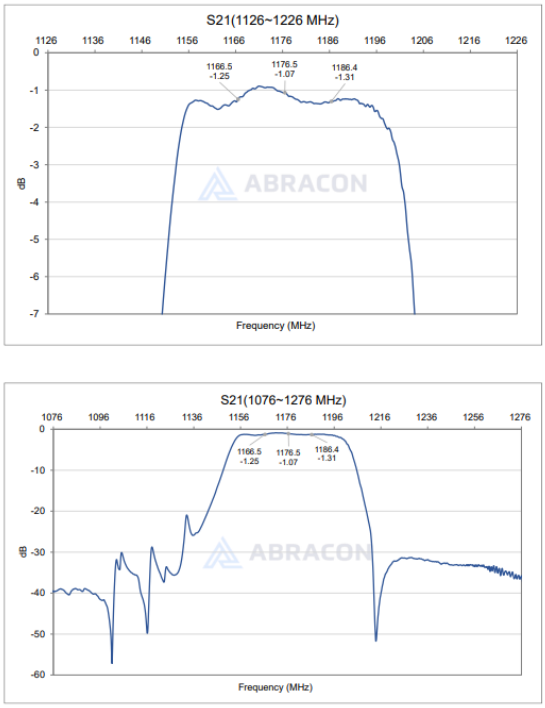

Figure 9.1.3– L5 SAW Filter Frequency Characteristics [24]

S21 narrow band

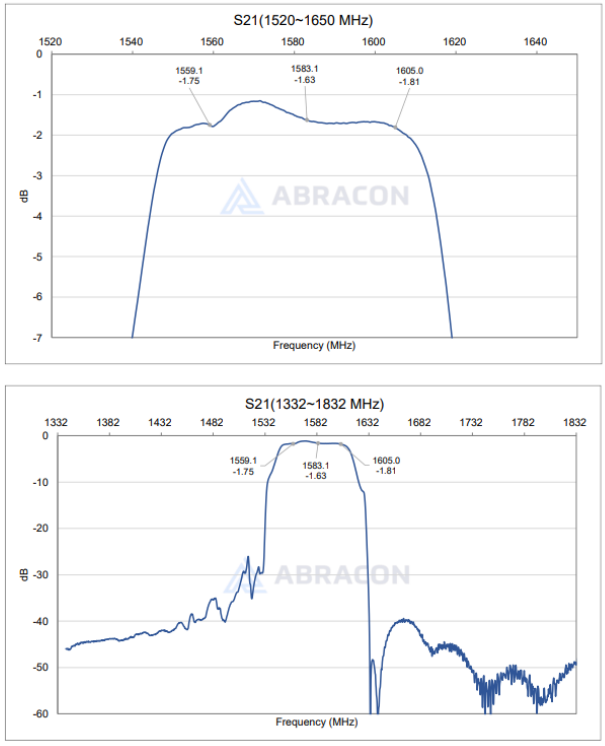

Figure 9.1.4 – L1 SAW Filter Frequency Characteristics [23]

9.2 Gantt Chart

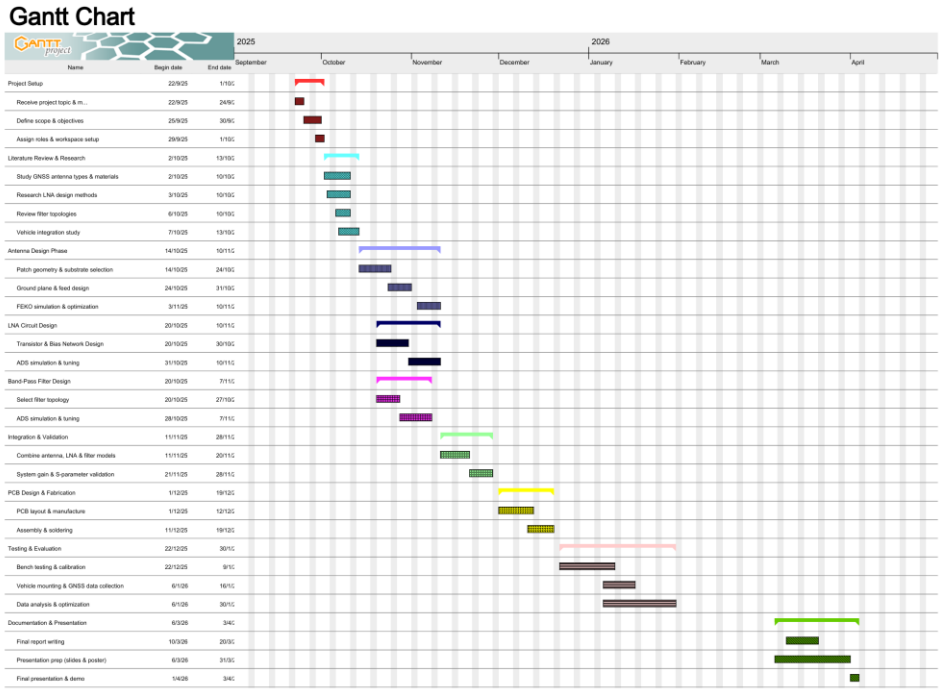

Figure 9.2.1 – Original Projected Gantt Chart

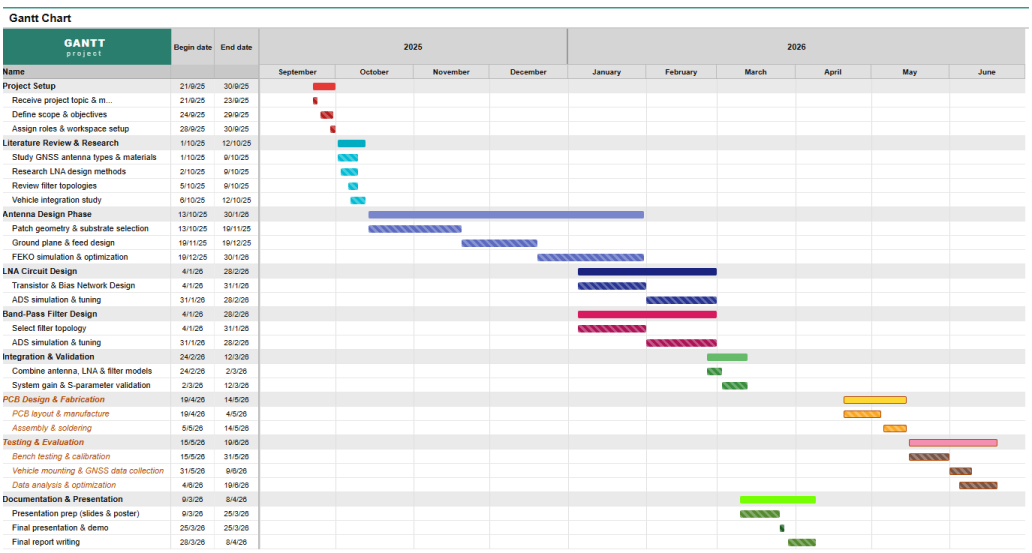

Figure 9.2.2 – Project's Actual Resulted Gantt Chart Timeline

The gap between the projected and actual timelines is not subtle, and most of it traces back to how much harder this project turned out to be than we expected. The antenna and circuit design phases ran long because we were, in a lot of cases, figuring things out from scratch. There is very little online about designing dual-band GNSS front ends at this level of integration. What

does exist is either surface-level or assumes RF expertise that none of us had coming in. So, the learning happened inside ADS, through failed simulations and re-tuned parameters, not by following any established guide. That alone pushed both the antenna work and the LNA and filter design well past their original end dates.

Keysight ADS made it worse. None of us had used it before this project. Getting the substrate definitions right, setting up Momentum EM simulations correctly, and understanding what the output was telling us took weeks. Weeks that were never budgeted in the original schedule. By the time each subsystem had working simulation results, and the PCB layouts were finalized, we were already behind.

The factor that hurt the most, though, is entirely outside our control. Our PCBs were ordered through JLCPCB and have not arrived due to manufacturing and international shipping delays. Every testing and validation task depends on having those boards. The simulation results across all subsystems look strong, but that does not change the fact that no physical testing has been possible, and the timeline reflects that reality.